

Detection of Externally-Influenced Routing via Inverse Markov-Chain Methods

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Abstract

Satellite-navigation and live-traffic services increasingly shape urban route choice by reallocating guided drivers around congestion. This weakens the revealed-preference assumption behind demand estimation, route-choice modelling and transport policy: observed traffic may reflect a mixture of intrinsic driver preferences and external routing recommendations. This dissertation develops a *partial-observation routing-influence filter* for that setting. Given a road graph, sparse observation stations and two traffic regimes, the filter fits one inverse Markov-chain model per regime and scores each trajectory with a log-likelihood ratio. The route-choice estimator is the partial-observation inverse Markov-chain method of Morimura, Osogami and Idé [1]; the contribution here is the two-regime filter, its trajectory-level scoring rule and its diagnostic tests. A compact primer on the mathematical objects and evaluation metrics is provided in section D.

The filter is evaluated in three tiers. Tier 1 reproduces Morimura et al.'s synthetic recovery benchmark, validating the estimator implementation. Tier 2 uses controlled SUMO simulation with labelled sat-navigation rerouting. On long trajectories the fitted detector reaches an area under the receiver-operating-characteristic curve (AUC) of 0.708, compared with a full-observation empirical 1-step reference of 0.801. Feature regularisation improves per-row chain recovery across five SUMO seeds, although its aggregate AUC lift is not robust. A regularised empirical 2-step test lifts AUC by about +0.09 to +0.10, indicating higher-order route memory not captured by the current 1-step model. Tier 3 applies the same filter to 10.5 million reconstructed real automatic-vehicle-identification trips from Xuancheng (China), using rush-hour versus off-peak and Labour-Day versus normal travel as unlabelled behavioural proxies while controlling origin-destination mix. On rush-vs-off-peak, the fitted 1-step detector remains at chance across all tested zone granularities, even where the full-observation reference retains a weak signal. On the stronger Labour-Day split, full observation gives a significant within-OD signal (AUC = 0.567), but multistart validation-selected fitting falls to chance (AUC = 0.502), and an

empirical 2-step audit does not improve the reference. The resulting contribution is a validated filter for controlled labelled settings and a clear account of its current limits on unlabelled real-world traffic.

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1 Notation

A consolidated reference for the mathematical objects used throughout this report. Notation follows Morimura et al. [1] where it does, and is extended in section 2.3 where required.

Table 1: Key symbols.

Symbol	Meaning
$\mathcal{X}, n$	Full state space (directed road edges); $n = \mathcal{X} $.
$X_o \subset \mathcal{X}$	Subset of <i>observed</i> states at which counts are available.
$\text{adj}(x)$	Set of legal successors of state x (out-neighbours in the directed graph).
$p_T(x' x)$	One-step transition probability from x to x' ; row x of the kernel P_T .
$p_I(x)$	Initial / teleport probability on $\mathcal{X}$; row vector.
$\beta \in [0, 1]$	Continuation probability per step; together with p_I defines the stationary mixture $P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top$.
π	Stationary distribution of P ; obtained from $\pi^\top P = \pi^\top$, $\mathbf{1}^\top \pi = 1$.
$f(x)$	Empirical visit count (or normalised frequency) at x . Observed only for $x \in X_o$.
$g(x, x')$	Empirical (or analytical) <i>hitting rate</i> from x to x' , a discounted first-passage probability on $X_o \times X_o$.
$\phi_T(x') \in \mathbb{R}^{d_T}$	Destination-state feature vector (section 2.3).
$\psi(x, x') \in \mathbb{R}^{d_\psi}$	Directed-edge feature vector.
$\nu \in \mathbb{R}^n$	Per-state base parameters of p_I .
$\omega^{\text{loc}} \in \mathbb{R}^E$	Per-edge local weights of p_T , with $E = \sum_x \text{adj}(x) $ the number of directed edges.
$\omega^{\text{gl}_1} \in \mathbb{R}^{d_T}$	Shared weights on destination features ϕ_T (paper Eq. 17).
$\omega^{\text{gl}_2} \in \mathbb{R}^{d_\psi}$	Shared weights on edge features ψ (paper Eq. 17).
θ	Full parameter vector $[\nu; \omega^{\text{loc}}; \omega^{\text{gl}_1}; \omega^{\text{gl}_2}] \in \mathbb{R}^d$, $d = n + E + d_T + d_\psi$.
λ	Tikhonov regularisation strength on $\ \theta\ ^2$.
$\gamma \in [0, 1]$	Mixing weight between the stationary-match loss L_d and the hitting-rate loss L_h .
$\Lambda(\tau)$	Log-likelihood-ratio score of a test trajectory τ under two fitted chains (section 2.3).
AUC	Area under the ROC curve of the LR score against labelled regime test trajectories.
$\text{RMAE}(\hat{\pi}, \pi)$	Relative mean absolute error of recovered stationary mass at unobserved states.

Throughout, “regime” refers to one of two distinguishable driver populations whose chains we fit and compare. In controlled simulation this is intrinsic vs. sat-navigation-influenced; in real-world data it is a behavioural proxy (e.g. rush-hour vs. off-peak, or Labour Day vs. normal days). The contrast $\Delta_{\text{fit}}(x, \cdot) = p_T^{(B)}(x, \cdot) - p_T^{(A)}(x, \cdot)$ between two fitted chains is the central quantity assessed against the empirical ground-truth contrast $\Delta_{\text{emp}}(x, \cdot)$.

2 Introduction

2.1 Context and research gap

Satellite-navigation and live-traffic services have become part of the route-choice system rather than passive map displays. Travel information changes route-choice behaviour in congested networks [2, 3], and routing-app shortest-path guidance can redistribute traffic in ways that affect wider network performance and residential cut-through traffic [4–6]. This creates a measurement problem. Traffic counts and trajectories no longer necessarily reveal only the preferences of unaided drivers; they may also encode the recommendations of external routing systems reacting to congestion and to other guided vehicles. This is a closed-loop data problem: route recommendations alter the driver behaviour from which future route-choice models may then be estimated [7]. It motivates looking for changes in local vehicular movement patterns, in the spirit of Shorten and co-authors’ junction-level pattern-identification work [8].

This matters because route-choice inference, demand estimation, pricing and infrastructure appraisal rely on revealed-preference data. If a substantial fraction of observed trips is externally guided, downstream models may mistake recommender-induced behaviour for intrinsic driver preference. The problem addressed here is therefore detection: given only sparse observation of a road network, can we distinguish trajectories that look intrinsic from trajectories that look externally influenced? The practical end goal is a filter: flag or remove the externally influenced trajectories so that a later analyst can pass a cleaned, intrinsic-behaviour dataset to downstream route-choice, demand-estimation or policy-appraisal tasks.

2.2 Related work and novelty

Route choice is often represented as a sequence of link-level decisions. Multinomial-logit and recursive-logit models parameterise the probability of taking the next road edge from the current edge [9–12]. In a one-step Markov-chain (MC) form, with road edges as states, the transition probability is

$$p_T(x' | x) = \frac{\exp(s(x, x'))}{\sum_{x'' \in \text{adj}(x)} \exp(s(x, x''))}, \quad (1)$$

where $\text{adj}(x)$ is the set of legal successors of edge x and $s(x, x')$ is a feature-based score (see sections D.1 and D.4 for the Markov-chain and softmax definitions used here). Markov-chain road-network dynamics have also been used by Shorten and co-authors as a compact representation for network-level regulation and control [13]. Related work also estimates route-choice models from incomplete trips, sparse sensor observations or inverse preference learning [14–16]. Those papers address important adjacent estimation problems, but they are normally framed as single-behavioural-model estimation rather than two-regime trajectory detection.

Morimura, Osogami and Idé introduced an inverse Markov-chain method for recovering a softmax-parametric chain from partial observations [1]. Their method fits a chain using counts at a subset of observed states X_o and, optionally, hitting rates between

observed states. It is the estimator used in this dissertation. The contribution here is not a new inverse-chain estimator. It is a two-regime partial-observation filter that fits one chain per regime, scores each trajectory with a log-likelihood ratio and reports diagnostics that separate transition-level chain recovery from aggregate confounds such as origin-destination (OD) mix, optimiser basins and model-order misspecification.

2.3 Filter specification and assumptions

The filter takes a directed road graph $\mathcal{X}$, an observed subset $X_o \subseteq \mathcal{X}$, regime-labelled station counts f_A, f_B and held-out trajectories. For each regime it fits a softmax-parametric MC ($\hat{P}_A$ and $\hat{P}_B$) by minimising Morimura et al.’s regularised log-ratio loss [1]. In the notation used here, the objective is

$$\begin{aligned} L(\theta) &= \gamma L_d(\theta) + (1 - \gamma) L_h(\theta) + \lambda R(\theta), \\ L_d(\theta) &= \frac{1}{2} \sum_{i,j \in X_o} \left(\log \frac{\pi_i(\theta)}{\pi_j(\theta)} - \log \frac{f_i}{f_j} \right)^2, \\ L_h(\theta) &= \frac{1}{2} \sum_{i,j \in X_o} (\log H_{ij}(\theta) - \log g_{ij})^2, \quad R(\theta) = \frac{1}{2} \|\theta\|_2^2. \end{aligned} \tag{2}$$

Here L_d matches observed station-count ratios through the model stationary distribution $\pi(\theta)$, L_h matches observed hitting rates g_{ij} through the model hitting rates $H_{ij}(\theta)$, γ controls the count-versus-hitting-rate balance, and λ is ridge regularisation. In the full-network SUMO and Xuancheng fits I mainly use the f -only setting ($\gamma = 1$), because empirical g is sparse and floor-dominated at city scale (section 4.3.7). All parametric fits use L-BFGS-B (section D.11), because the objective is smooth, analytic gradients are available, and the parameter dimension is large enough that a limited-memory quasi-Newton method is a practical compromise between first-order gradient descent and full second-order Newton optimisation. The fitted chains $\hat{P}_A$ and $\hat{P}_B$ then define a log-likelihood-ratio (LLR) score

$$\Lambda(\tau) = \sum_t [\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)],$$

using the log-likelihood ratio because it is the natural ranking statistic for deciding which of two fitted generative models better explains a trajectory. Under correctly specified regime models, thresholding the likelihood ratio is the optimal binary test in the Neyman–Pearson sense [17], and in the Markov setting the score decomposes additively into interpretable per-transition evidence. Alternative scalar scores were considered but rejected as the primary score: posterior probabilities require regime priors and calibration, single-chain likelihoods do not directly compare the two hypotheses, transition-distance scores discard the full generative path probability, and discriminative classifiers require labelled training trajectories that the real Xuancheng tier does not have. The LLR therefore gives the cleanest common score across the labelled SUMO and unlabelled-proxy Xuancheng tiers. Its threshold can be adjusted for a desired false-positive/false-negative tradeoff without refitting either chain. Validation reports the area under the receiver-operating-characteristic curve (AUC) over possible decision thresholds (sections D.5 and D.6).

The Morimura estimator carries assumptions that are used later in the report. First, it assumes a one-step MC: the next edge depends on the current edge, not the full previous path. Second, its count term treats f as a sample from a stationary distribution, and its optional hitting-rate term treats g as stationary first-passage evidence. Third, the softmax form restricts recoverable behaviour to the chosen feature and parameter family. In the SUMO and Xuancheng tiers the data come from finite, time-varying traffic windows, so the fitted chain is a time-aggregated surrogate for that window rather than behavioural ground truth. The empirical full-observation transition matrix used for comparison is also a finite-data reference, not an oracle.

Diagnostics. The scalar AUC is reported with transition-level diagnostics. At branching states, the fitted regime contrast $\Delta_{\text{fit}} = \hat{P}_B - \hat{P}_A$ is compared with the full-observation empirical contrast Δ_{emp} using stacked Pearson correlation and per-state cosine similarity (sections D.7 and D.8). A high AUC with a near-zero transition contrast indicates aggregate-bias-driven detection, whereas positive contrast correlation supports a chain-recovery claim.

Extensions. Three extensions are evaluated in the tiers. The feature-regularised variant adds road class, speed, lane and turn-geometry features as shared parameters. The empirical two-step test checks whether (x_{t-1}, x_t) -conditional routing memory improves over the one-step reference before a parametric two-step inverse-MC model is attempted. The OD-rebalancing step (section D.9) matches real-world regime pools by (origin zone, destination zone) before comparing rush-hour and off-peak routing.

Validation split. Test trajectories are withheld from the empirical transition matrix and from the empirical-reference LLR scorer. The inverse-MC fit uses aggregate station counts for the whole regime window, not counts recomputed after removing held-out vehicles. The reported AUCs are therefore held out at the trajectory-scoring level, not a fully out-of-sample sensor-count experiment.

2.4 Experimental structure

The report follows the experimental tiers directly. Tier 1 validates the inverse-MC estimator against Morimura et al.’s synthetic benchmark. Tier 2 uses Simulation of Urban MObility (SUMO) to test the full filter under labelled sat-navigation rerouting. Tier 3 applies the same filter to automatic vehicle identification (AVI) data from Xuancheng, China, where rush-hour versus off-peak and Labour-Day versus normal travel are used as unlabelled behavioural proxies for external routing influence. The Discussion compares what generalises across tiers, the Conclusion states the supported claims, and the final section covers project management and reflection.

3 Tier 1: Validating the inverse-MC estimator inside the filter

3.1 Motivation

This tier validates the estimator component of the filter in isolation, before either the scoring rule or the regime-detection layer is attached. If the inverter cannot reproduce the published recovery curve on its native synthetic benchmark, no downstream filter result on SUMO or Xuancheng is meaningful. We therefore replicate the §6.1 synthetic experiment of Morimura et al. [1] end-to-end and verify that we reach the published relative mean absolute error (RMAE) curve to within ~ 0.05 across observation fractions in $[0.10, 0.90]$.

3.2 Method

A synthetic chain is constructed on $n = 100$ states with a random strongly-connected adjacency, mean out-degree ~ 3 . The true transition kernel is a softmax of eq. (1) with all parameters $(\nu, \omega^{\text{loc}}, \omega^{\text{glo}_1}, \omega^{\text{glo}_2})$ drawn iid $\mathcal{N}(0, 1)$, with $d_T = d_\psi = 5$ random global features per the paper. The truth is then mixed 70/30 with a Dirichlet(0.3) noise row-distribution to push it slightly outside the parametric family ($\sim 4\%$ TV distance) and so present the inverter with a model-class gap analogous to a real-world scenario.

For each observation fraction $|X_o|/n \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.70, 0.90\}$, X_o is randomly sampled, and the inverter is fitted on the analytic f and g derived from the true chain at X_o , with Tikhonov regularisation λ cross-validated over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ on a 75/25 split of X_o . The reported metric is the relative mean absolute error

$$\text{RMAE}(\hat{\pi}, \pi) = \frac{\frac{1}{|X \setminus X_o|} \sum_{x \notin X_o} |\hat{\pi}(x) - \pi(x)|}{\frac{1}{n} \sum_{x \in X} \pi(x)}, \quad (3)$$

i.e. the mean absolute error on unobserved states normalised by the average true stationary mass. The recovered $\hat{\pi}$ is rescaled by a scale-free convention discussed in section 3.4.1. Each $|X_o|$ value is repeated for 10 trials with fresh truth + X_o draws, matching the visible error-bar count in the original Fig. 2A.

3.3 Results

The reproduction RMAE curve, our “Proposed (with g)” configuration, tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 90]$. table 2 summarises.

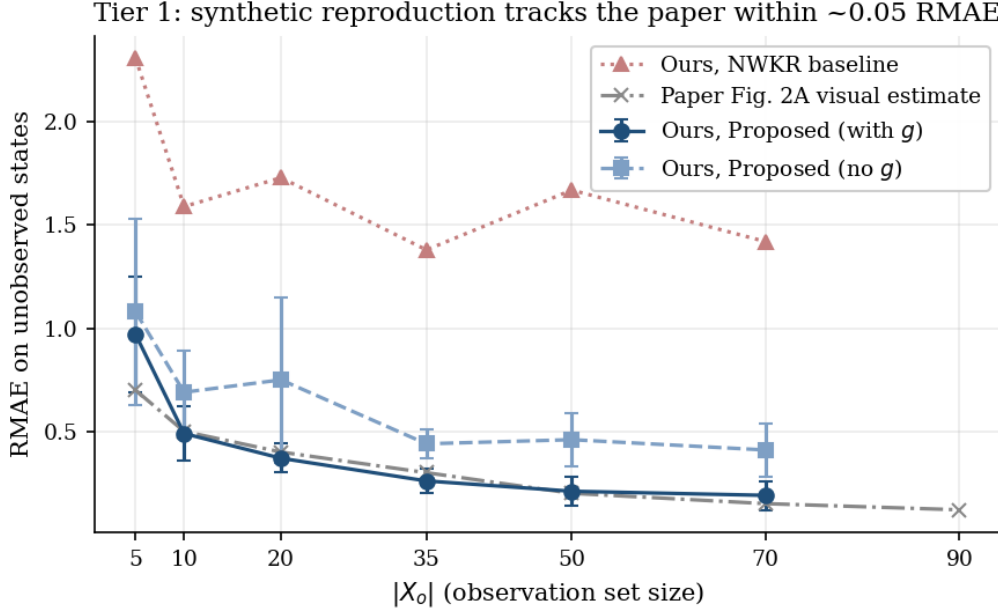


Figure 1: Tier 1 synthetic reproduction tracks the paper’s Fig. 2A within ~ 0.05 RMAE across $|X_o| \in [10, 70]$. Our “Proposed (with g)” (dark blue) is the headline curve and matches the paper (grey) closely; “Proposed (no g)” (light blue) is the f-only ablation; NWKR (red, dotted) is the comparator baseline. Error bars are ± 1 standard deviation across 10 trials.

Table 2: Reproduction RMAE vs. visual estimates from paper Fig. 2A. Mean over 10 trials, $n = 100$, full §6.1 protocol including global features and cross-validated λ . Same data as fig. 1.

$ X_o $	Proposed (with g)	Proposed (no g)	NWKR baseline	Paper Fig. 2A visual estimate
5	0.97 ± 0.28	1.08 ± 0.45	2.31	~ 0.7
10	0.49 ± 0.13	0.69 ± 0.20	1.59	~ 0.5
20	0.37 ± 0.07	0.75 ± 0.40	1.73	~ 0.4
35	0.26 ± 0.06	0.44 ± 0.07	1.38	~ 0.3
50	0.21 ± 0.07	0.46 ± 0.13	1.67	~ 0.2
70	0.19 ± 0.07	0.41 ± 0.13	1.42	~ 0.15

The single value where our number is materially off the paper’s is $|X_o| = 5$, where our $\text{RMAE} \approx 0.97$ against the paper’s ~ 0.7 . Investigation revealed this is a λ -CV degeneracy: with $|X_o| = 5$ states, the 75/25 split puts ≈ 1 state in the validation set, which is not enough to discriminate λ values robustly, and the cross-validated λ over-regularises. The Nadaraya–Watson kernel-regression baseline (“NWKR”) does not drop as steeply as the paper’s, which is most plausibly a comparator- implementation mismatch rather than a failure of the inverse-MC implementation.

3.4 Interpretation

3.4.1 Prediction rescaling convention

The paper writes the recovered prediction as $\hat{f}(x) = \hat{c} \hat{\pi}(x)$ with $\hat{c} = \overline{f}_{X_o}$ (their §6.1). Taken literally this collapses every RMAE to ≈ 1 because `stationary()` returns a

probability summing to 1 over n states (so $\hat{\pi}(x) \sim 1/n$ while $f(x) \sim K/n$), under-scaling by a factor of n . The operationally correct convention, and the one that yields the paper’s reported numbers, is the scale-free form $\hat{f}(x) = (\sum_{X_o} f) \hat{\pi}(x) / (\sum_{X_o} \hat{\pi})$, which reduces to the literal formula when $\sum_{X_o} \hat{\pi} = 1$. We verified this empirically: applying the literal formula collapses all RMAE values to ≈ 1.0 . The scale-free form is preserved throughout the rest of this work.

3.4.2 Gradient verification

Across three random parameter draws including local-only, ϕ_T -only, and ψ -only edge-case configurations, the centred finite-difference gradient check returns maximum relative error $\leq 5 \times 10^{-9}$ for every block, confirming the analytic derivation. Details of the finite-difference (FD) check infrastructure are in `morimura.py`.

3.4.3 Headline outcome

The synthetic reproduction is locked in as a faithful reference. The proposed-method curve matches the original benchmark closely for $|X_o| \geq 10$, while the $|X_o| = 5$ case (a λ -CV degeneracy) and the NWKR comparator (an implementation gap, not a method error) are documented deviations. With those two caveats, the curve shape and method ordering match the paper. This implementation serves as the upstream operational reference for all subsequent validation tiers; downstream limitations in Tier 2 and Tier 3 are attributable to the data and the model class at scale, not to the upstream framework as implemented here.

4 Tier 2: Controlled validation of the filter in SUMO

4.1 Motivation

This tier is the controlled, end-to-end validation of the partial-observation routing-influence filter in the only setting that supplies ground-truth regime labels: an identical population of vehicles is simulated once with sat-navigation rerouting turned off and once with it turned on. Every component of the filter is exercised at city-block scale ($n = 2,405$ edges): the inverse-MC estimator is fitted per regime, the log-likelihood-ratio scorer is applied to held-out trajectories, the feature-regularised filter variant is tested against a no-features baseline, and the filter’s diagnostic suite (per-row Pearson/cosine, multistart basin check, empirical 2-step memory audit) is run on the resulting fits. A trajectory-length sweep traces how the scoring window T shifts both the empirical reference and the fitted detector together, and we close with a documented infeasibility of the hitting-rate ($f + g$) extension at this scale, a design-relevant failure mode for the filter’s optional g input.

A note on terminology used throughout this section. We refer to the no-rerouting regime as “static unguided routing”: SUMO routes those vehicles by static shortest-path Dijkstra with no live rerouting. This is a controlled proxy for intrinsic route choice, not a behavioural model of human preference, and we interpret the fitted “intrinsic” chain accordingly. We also use “empirical 1-step reference” (rather than an information-theoretic ceiling) for the AUC obtained from the full-observation Laplace-smoothed transition matrix; it is the best a 1-step detector can do *given this finite data and smoothing*, and finite-sample and train/test variation can move it (and occasionally produce apparent

anomalies, section 5.4.1).

4.2 Method

4.2.1 Scenario construction

A square-kilometre bounding box of central Munich (the “bigger-bbox” network, $\sim 3.7\text{ km} \times 3.3\text{ km}$, derived from OpenStreetMap (OSM) via `netconvert`) yields $n = 2,405$ directed edges after filtering to the largest strongly-connected passenger-allowed component, with $E = 5,987$ directed transitions and mean out-degree ~ 2.5 . A demand of $\sim 28,800$ vehicles is generated by `randomTrips.py` over a four-hour horizon at $-p0.5$. Each vehicle is replayed twice under the same insertion times and origin/destination pairs:

- **Intrinsic regime:** `-device.rerouting.probability 0`. Vehicles route via static shortest-path Dijkstra and never react to congestion.
- **Sat-navigation regime:** `-device.rerouting.probability 1`. Every vehicle re-evaluates its route every 30s on the basis of observed live edge travel times.

Operating at $-p0.5$ was selected from preliminary demand sweeps: at higher demand both regimes saturate and the sat-navigation contrast collapses; at lower demand there is insufficient congestion to drive any rerouting. At $-p0.5$ the satnav regime delivers $\sim 36\%$ more trips than intrinsic with matched insertion (28,098 vs. 20,700 finishing). This completion imbalance is a limitation of the SUMO operating point. The validation scorer uses balanced test sets ($N = 300$ per regime), and the Morimura L_d term is scale-free in the aggregate counts, so the result is not a trivial class-prior effect. Nevertheless, the fitted counts and the pool of long completed trajectories are conditional on different completion processes in the two regimes. We therefore treat Tier 2 as a controlled operational validation of the filter under this scenario, not as a completion-balanced causal estimate of the effect of sat-navigation rerouting. A stricter sensitivity study would rerun the simulation or rebuild f under matched completed-trip populations; that extension is left outside the present experimental budget.

For each regime we expose vehicle counts `edgedata.xml` (the empirical f) and complete per-vehicle routes `vehroutes.xml` (used as the test set and to construct the empirical chain for the contrast diagnostic). The inverter sees only $|X_o| = 601$ randomly sampled edges, 25% of the network, consistent with the operating point of the Morimura §6.1 benchmark.

A stationarity caveat applies here and carries through to Tier 3. The Morimura observation model assumes the counts f are samples of a *stationary* distribution and that g are stationary hitting rates. SUMO produces finite-horizon, dynamic, congestion-affected counts over a four-hour window, so the `edgedata` totals are best treated as empirical occupancy / visit-frequency summaries rather than exact stationary samples. We treat them as such: the inverse fit is a controlled operational test of the pipeline under a stationarity approximation, not a claim that the fitted chain is behavioural ground truth. This is one reason we report the full-observation empirical transition matrix as a finite-data reference rather than as a true ceiling, and why the per-row contrast diagnostic (section 2.3), which compares fitted and empirical *contrasts* on the same finite data, is the load-bearing evidence rather than the absolute AUC.

4.3 Results

4.3.1 Single-seed result: features and the inverse fit

We compare `-features none` (local-only fit, $d = n + E = 8,392$) to `-features real` (local + global feature terms, $d = 8,402$). On seed=42, $T = 20$, $|X_o| = 601$, $\text{maxiter} = 1500$, the headline numbers are summarised in table 3.

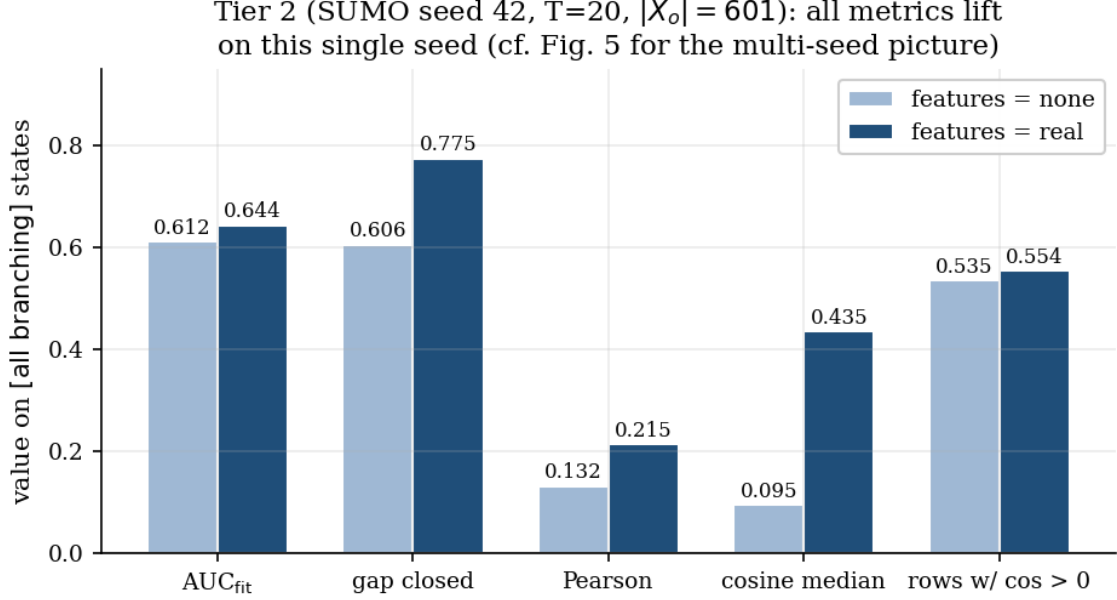


Figure 2: Tier 2 features lift on SUMO bigger-bbox, seed 42, $T = 20$. Every metric, including detection AUC, gap-closed, per-row Pearson, per-row cosine median and fraction of rows with positive cosine, moves positively under `features=real` (dark) vs. `features=none` (light). The Pearson +63% relative lift and the cosine median 4.6 $\times$ lift on the [all branching] generalisation block are the strongest evidence that chain recovery, not aggregate-bias drift, is what features improve.

Table 3: Features lift across all metrics on the SUMO bigger-bbox, seed=42 (fig. 2). [all branching] restricts the per-row diagnostic to states with out-deg > 1 that the inverter did *not* directly observe, namely the generalisation block.

metric ([all branching])	features=none	features=real	Δ
AUC _{emp} (full-obs. reference)	0.685	0.685	–
AUC _{fit}	0.612	0.644	+0.032
gap closed	60.6%	77.5%	+16.9 pp
stacked Pearson($\Delta_{\text{fit}}, \Delta_{\text{emp}}$)	+0.132	+0.215	+63% rel
per-row cosine median	+0.095	+0.435	4.6 $\times$
rows with cosine > 0	53.5%	55.4%	+1.9 pp

Two results are worth separating. First, the +0.032 AUC lift on this seed sits outside the single-instance variability measured in repeated SUMO baseline runs. As the multi-seed analysis of section 4.3.2 shows, however, this aggregate AUC lift does *not* survive replication across SUMO instances. The per-row chain-recovery improvement, not the

aggregate AUC, is the robust effect. Second, the stacked Pearson on `[all branching]` rises from $+0.132$ to $+0.215$, and the per-row cosine median from $+0.095$ to $+0.435$. These are the metrics that distinguish “the fit improved because the chains got more accurate per-row” from “the fit improved because the chains drifted in helpful opposite directions in aggregate”. Both moved positively, supporting the stronger chain-recovery claim. Unlike the aggregate AUC, this per-row lift replicates across seeds (section 4.3.2).

The one metric that barely moves is the fraction of rows with positive cosine (only $+1.9$ pp on `[all branching]`, 58.6% at `[X_o branching]`). Features sharpen the contrast magnitude in the rows that were already approximately right, but do not flip the sign on the $\sim 45\%$ of branching states whose fitted contrast points the wrong way. This suggests the residual misspecification is structural to the 1-step softmax family. Section 4.3.4 tests that hypothesis.

4.3.2 Multi-seed validation: chain recovery is robust, detection AUC is not

To test whether the seed-42 lift replicates, we re-ran the contrast-correlation diagnostic across five independent SUMO instances (`randomTrips` seeds $\in \{7, 23, 42, 101, 2024\}$, X_o seed fixed at 13, $T = 20$, $|X_o| = 601$, `maxiter` = 1500) under both `features=none` and `features=real` (table 4, fig. 3).

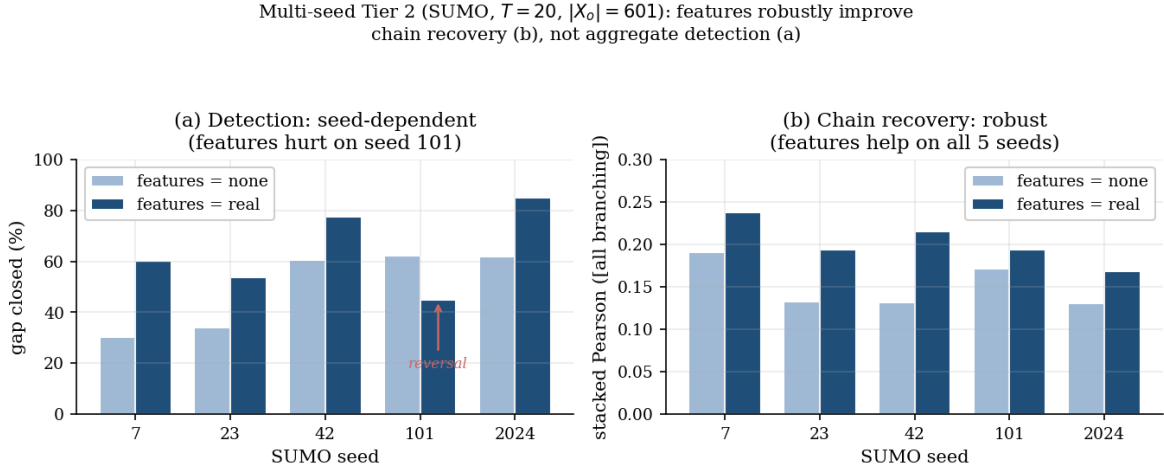


Figure 3: Multi-seed Tier 2 features sweep (five SUMO seeds, `[all branching]` block, $T = 20$). **(a)** The aggregate detection metric (gap-closed) lifts under `features=real` on four seeds but *reverses* on seed 101; the `features=none` bars also reproduce the bimodal basin structure ($\{7, 23\}$ low, $\{42, 101, 2024\}$ high). **(b)** The per-row chain-recovery metric (stacked Pearson) lifts on *all five* seeds. Features are a robust chain-recovery lever, not a robust aggregate-detection lever.

Table 4: Multi-seed features sweep on SUMO bigger-bbox (fig. 3), [all branching] block, $T = 20$, $|X_o| = 601$. The AUC lift reverses on seed 101; the Pearson lift is positive on all five seeds. Mean Δ rows are **real** – **none** with sample standard deviation.

seed	gap closed		AUC _{fit}		Pearson	
	none	real	none	real	none	real
7	30.3%	60.3%	0.544	0.588	+0.191	+0.238
23	34.2%	53.8%	0.554	0.585	+0.133	+0.194
42	60.6%	77.5%	0.612	0.644	+0.132	+0.215
101	62.1%	45.0%	0.581	0.559	+0.171	+0.194
2024	61.8%	85.1%	0.585	0.618	+0.131	+0.168
mean Δ	–		$+0.024 \pm 0.026$		$+0.050 \pm 0.023$	

Three findings. First, the aggregate detection lift is *not* statistically robust: the paired AUC_{fit} lift averages $+0.024 \pm 0.026$ across the five seeds, positive on four but reversing on seed 101 (a paired t -test gives $t \approx 2.0$, $p \approx 0.11$, not significant at $\alpha = 0.05$). The single-seed $+0.032$ headline of section 4.3.1 is therefore a favourable draw, not a replicable property of the aggregate AUC.

Second, the per-row chain-recovery metrics *are* robust. The stacked Pearson lift averages $+0.050 \pm 0.023$ and is positive on *all five* seeds ($t \approx 4.9$, $p \approx 0.008$); the per-row cosine median lifts on all five as well (mean $\approx +0.25$). Seed 101 illustrates the illustration of the AUC $\leftrightarrow$ chain-quality decoupling: features *hurt* its detection AUC ($0.581 \rightarrow 0.559$) while *improving* its chain recovery (cosine median $0.062 \rightarrow 0.475$). The features lever is therefore a **chain-recovery** improvement, not an aggregate-detection improvement.

Third, the **features=none** fit reproduces the bimodal gap-closure structure observed in repeated SUMO runs: seeds $\{7, 23\}$ cluster low (30–34%) and seeds $\{42, 101, 2024\}$ cluster high (61–62%), with nothing in between. The same seeds fall in the same clusters as the earlier run, indicating the split is a stable property of the L-BFGS basin structure (section D.11) on these instances rather than measurement noise. The **features=none** AUC_{fit} across seeds is 0.575 ± 0.027 , consistent with the baseline variability. Whether this low basin is escapable by multi-start re-initialisation, a question left open by the multi-seed sweep, is settled in section 4.3.3: it is not.

4.3.3 Multistart basin check: the low basin is not an initialisation artefact

The bimodal basin structure of section 4.3.2 raises an optimiser-quality question: is the low-gap basin on seeds $\{7, 23\}$ an artefact of the zero-initialisation that limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) starts from, escapable by multi-start re-initialisation? To test this we re-ran both low-basin seeds with $K = 5$ random restarts per chain (initial $\theta_0 \sim \mathcal{N}(0, I)$, **features=none**, $T = 20$), selecting the restart with the highest held-out validation log-likelihood and comparing it to the zero-init baseline (fig. 4).

Tier 2 multistart basin check (SUMO, features=none, $K = 5$, $T = 20$): the low basin is not an initialisation artefact – zero-init is the best optimum found

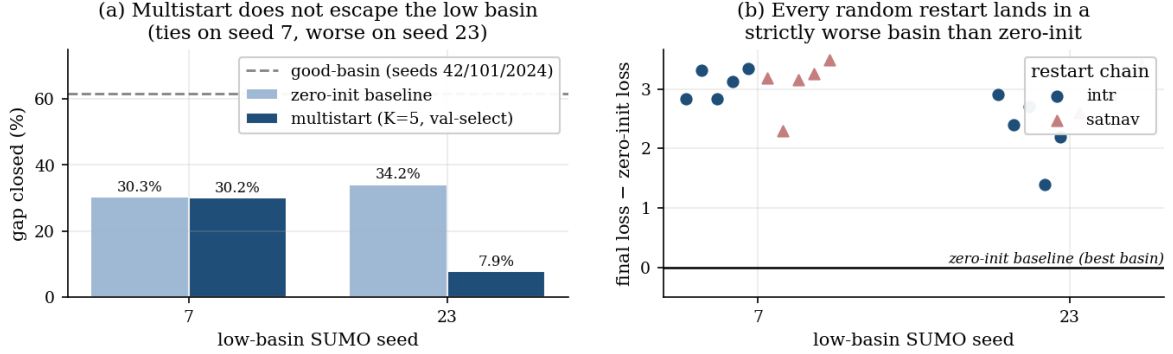


Figure 4: Tier 2 multistart basin check on the two low-basin seeds ($\text{features}=\text{none}$, $K = 5$, $T = 20$). **(a)** The val-selected multistart detector ties the zero-init baseline on seed 7 and is markedly worse on seed 23; neither approaches the $\sim 61\%$ good-basin level of seeds $\{42, 101, 2024\}$ (dashed). **(b)** Every one of the ten random restarts (five per regime) converges to a strictly *higher* final loss than zero-init. Zero-init is therefore the best optimum found, and the low basin is not an initialisation artefact.

The answer is negative, and informatively so. On *both* seeds, every one of the ten random restarts (five per regime) converged to a strictly *worse* training loss *and* a worse held-out validation log-likelihood than the zero-init baseline (fig. 4b): on seed 7 the restarts reach intrinsic-chain loss 44.99–45.5 against the baseline’s 42.15; on seed 23, 457–459 against 455.9. The val-selected multistart detector therefore *ties* the baseline on seed 7 (30.2% vs 30.3% gap closed) and is markedly *worse* on seed 23 (7.9% vs 34.2%); neither approaches the $\sim 61\%$ good-basin level of seeds $\{42, 101, 2024\}$ (fig. 4a).

Two conclusions follow. First, the cross-seed bimodality is *not* an initialisation artefact: zero-init is the best optimum found on these instances and random restarts cannot reach the high basin, so the low gap on seeds $\{7, 23\}$ is a property of those instances’ loss landscapes rather than of where the optimiser starts. This closes the open question left by section 4.3.2 and reinforces the conclusion that the split is structural. Second, the seed-23 collapse to 7.9% is itself a small cautionary result: selecting the fit by held-out *log-likelihood* picked a chain that *discriminates worse* (the val-selected sat-nav restart was among the lowest-AUC of the five), another instance of the likelihood $\leftrightarrow$ detection decoupling that recurs throughout this work (section 4.3.1). The caveat is that all fits were `maxiter`-capped at 1,500 iterations and the random inits have large norm ($\|\theta_0\| \approx 92$ at $d = 8,392$); we have thus shown that the *standard* random-restart recipe does not escape the low basin, not that the basin is provably global.

4.3.4 Higher-order memory diagnostic: a regularised 2-step empirical lifts the reference

A natural hypothesis for the still-flipped rows is that the 1-step Markov assumption is empirically violated: real drivers’ next-edge choice depends on *both* the current and previous edge (“do not U-turn”, “keep on the arterial”), and the 1-step softmax cannot encode that dependence. If true, an empirical 2-step detector that uses (x_{t-1}, x_t) as context

should out-discriminate the 1-step empirical. The question is whether any reasonably-regularised 2-step empirical can lift the reference, given that the raw (x_{t-1}, x_t) pair space is huge ($|X|^2 \approx 5.8\text{M}$ at the bigger-bbox scale) and most of it is empty.

We test two regularisation schemes across the same five SUMO seeds used in section 4.3.2 (training/test split unchanged):

Dirichlet smoothing (toward uniform-over-neighbors). $\hat{P}_{2,\alpha}(y|x_p, x) = (c(x_p, x, y) + \alpha) / (c(x_p, x) + \alpha \cdot |\text{adj}(x)|)$, swept over $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$.

Interpolation backoff (toward 1-step empirical). $\hat{P}_{\text{back}}(y|x_p, x) = \lambda \cdot \hat{P}_2^{\text{MLE}}(y|x_p, x) + (1 - \lambda) \cdot \hat{P}_1(y|x)$, where $\hat{P}_2^{\text{MLE}}$ is the unsmoothed 2-step MLE and $\hat{P}_1$ is the 1-step empirical chain that elsewhere serves as the full-observation reference. Two λ modes: fixed $\lambda \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ and count-conditional $\lambda = N(x_p, x) / (N(x_p, x) + \kappa)$ with $\kappa \in \{1, 10, 100, 1000\}$. The count-conditional mode trusts the 2-step MLE in proportion to its observation support, falling back smoothly to the 1-step empirical for rare contexts.

Table 5: 2-step empirical sweep on SUMO bigger-bbox across 5 SUMO seeds ($T = 20$, training/test split as in section 4.3.2, 1-step empirical baseline mean 0.652 ± 0.022 across these seeds). Each row reports the per-method best multi-seed configuration with its mean and standard deviation across the five seeds. The “single-seed (seed 42)” row gives the single-seed comparison point; the multi-seed rows below are the load-bearing evidence.

method	best config	AUC _{2step}	Δ vs 1-step	seeds winning
single-seed Dirichlet (seed 42)	$\alpha \rightarrow 0$	0.666	−0.019	0/1
Dirichlet α (multi-seed)	$\alpha = 10^{-3}$	0.673 ± 0.012	+0.021	4/5
backoff fixed- λ (multi-seed)	$\lambda = 0.9$	0.747 ± 0.012	+0.095	5/5
backoff count-conditional (multi-seed)	$\kappa = 100$	0.752 ± 0.015	+0.100	5/5

Three findings (table 5).

First, the apparent *negative* result from a single-seed Dirichlet analysis ($0.666 < 0.685$ at seed 42, $\Delta = -0.019$) is driven by two compounding errors. Seed 42 has the highest 1-step AUC of the five SUMO instances (0.685 vs. mean 0.652), leaving the least room for any 2-step lift; multi-seed averaging exposes that the seed-42 ceiling masked. Separately, Dirichlet shrinks the 2-step estimate toward uniform-over-neighbors, which discards the marginal routing information that the 1-step empirical already encodes. The natural shrinkage target is the 1-step empirical itself, not uniform.

Second, interpolation backoff toward the 1-step empirical resolves both issues. The count-conditional weight $\lambda(x_p, x) = N(x_p, x) / (N(x_p, x) + \kappa)$ trusts the 2-step MLE proportionally to its observation support; at $\kappa = 100$ the per-seed Δ vector is $\{+0.090, +0.094, +0.099, +0.107, +0.101\}$ (s42, s23, s2024, s7, s101), mean +0.100, standard deviation 0.009. Every seed wins, and the lift is tighter than any other multi-seed effect in this section (section 4.3.2’s features Pearson lift was $+0.050 \pm 0.023$ on the per-row metric; the aggregate AUC lift $+0.024 \pm 0.026$ reversed on seed 101). At $\kappa = 100$ the effective λ across the 3,528 observed (x_p, x) pairs on seed 42 has median 0.21, 10th–90th percentiles $[0.03, 0.81]$: the

detector mostly relies on the 1-step baseline and only trusts the 2-step part where strong observation support exists.

Third, the lift is not knife-edge in the hyperparameter. The fixed- λ sweep gives $\Delta \in \{+0.088, +0.092, +0.094, +0.095\}$ for $\lambda \in \{0.25, 0.5, 0.75, 0.9\}$, all positive across all 5 seeds. The count-conditional sweep gives $\Delta \in \{+0.095, +0.097, +0.100, +0.097\}$ for $\kappa \in \{1, 10, 100, 1000\}$. The hyperparameter selection caveat applies in the narrow technical sense ($\kappa = 100$ was the sweep maximum on the same test set used for AUC, so the exact $+0.100$ is mildly test-tuned) but the qualitative conclusion is robust to selection: *any reasonable backoff strength lifts AUC by $\sim +0.09$ to $+0.10$ across all five SUMO seeds.*

Implication. A properly regularised 2-step empirical *does* lift the reference, decisively, across every SUMO seed tested: the regularised 2-step AUC reaches 0.752 ± 0.015 against the 1-step baseline of 0.652 ± 0.022 , a $+0.100$ delta whose per-seed signs all point the same way. Two implications follow.

The per-row chain-recovery gap left by the 1-step features lever (section 4.3.1: $+0.215$ stacked Pearson on [all branching], $\sim 45\%$ of rows still with wrong-direction contrast) is consistent with the 1-step softmax family being structurally unable to express (x_{t-1}, x_t) -conditional choice. This is the regime where a parametric 2-step Morimura extension is the right design lever, with the regularised empirical sitting roughly $+0.10$ AUC above the 1-step ceiling at $T = 20$ as the performance target it would need to approach.

We do *not* report an AUC_{fit} for the regularised 2-step here. A parametric 2-step Morimura inverter is a non-trivial extension with a different state space, different gradient and different ceiling argument, and is left to future work (section 6.4). The empirical result above establishes that the signal exists and bounds the achievable 2-step inverter performance from above; what it does not do is settle whether the parametric inversion route can recover that bound under partial observation.

4.3.5 Trajectory-length sweep: lifting the ceiling and the detector together

The 0.685 ceiling is specific to the chosen test trajectory length $T = 20$. The log-likelihood-ratio score $\Lambda(\tau)$ is a sum of T independent log-ratio increments, so its variance scales as $\sim T$. Longer trajectories produce a tighter score distribution between the two regimes and a higher achievable AUC; the data is “the same”, only the observation window is wider.

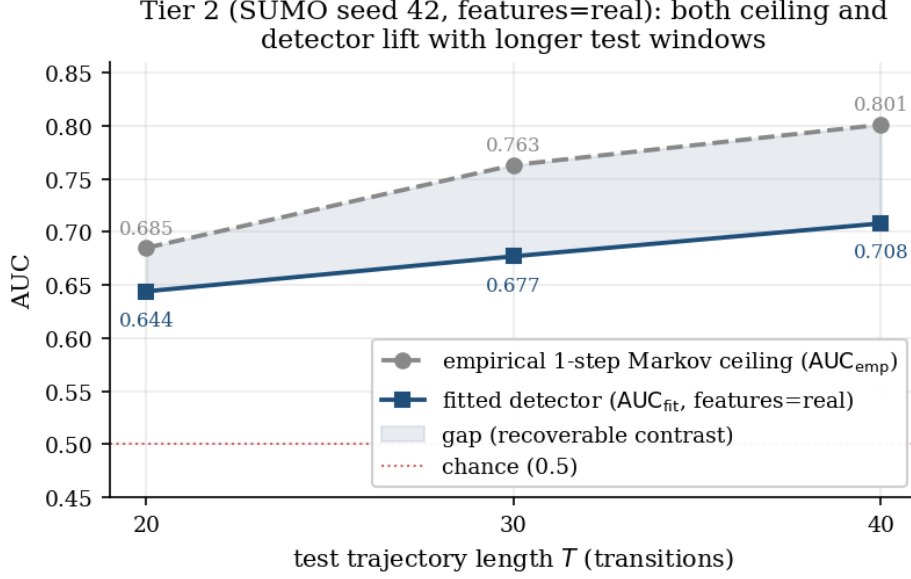


Figure 5: Tier 2 T -sweep on SUMO bigger-bbox, seed 42, **features=real**. The empirical 1-step Markov ceiling (grey dashed) rises with T as expected for a variance-shrinking score; the fitted detector (blue solid) tracks it with roughly constant gap. The shaded region is the recoverable contrast not yet captured by the fitter. The fitted chain itself is T -independent (identical final losses across runs); only the trajectory-level scoring window changes.

Table 6: T -sweep on SUMO bigger-bbox, seed=42, **features=real** (fig. 5). The fit itself is T -independent: only the test scoring window changes, as confirmed by the identical final-loss values 836.4/529.4 across all three runs.

T	AUC_{emp}	AUC_{fit}	gap closed	Pearson (all-br)
20	0.685	0.644	77.5%	+0.215
30	0.763	0.677	67.4%	+0.220
40	0.801	0.708	69.0%	+0.216

Two findings worth recording. First, the empirical ceiling rises from 0.685 to 0.801 between $T = 20$ and $T = 40$, confirming that the ceiling at $T = 20$ was an artefact of the truncation window, not a fundamental data limit. Second, the per-row Pearson is constant across T at $\sim +0.21$. T is purely a scoring-window lever: it lets the trajectory-level score discriminate more sharply, but it does not change the chain that was fit. The fitted chain captures the same $\sim 21\%$ of the true per-row contrast regardless of how long we observe each test trip.

The strongest single-seed SUMO headline is therefore $AUC = 0.708$ at $T = 40$, 69% of the gap closed against the full-observation empirical 1-step reference of 0.801, with a stable per-row chain Pearson of +0.216 on [all branching].

4.3.6 Disentangling sat-nav adoption from congestion-induced rerouting

A methodological concern arises in advance of Tier 3 (section 5), where no ground-truth sat-navigation label exists and we will rely on a behavioural rush-vs-off-peak proxy. Rush hour and off-peak differ in *two* ways simultaneously: in sat-navigation adoption

rate (presumably higher when journeys are time critical) and in congestion level (heavy demand changes routing incentives for *all* drivers, sat-nav users and unaided drivers alike). Real data cannot separate the two contributions, because there is no Xuancheng configuration in which the congestion level is varied while the sat-navigation adoption rate is held fixed (or vice versa). SUMO can: the per-vehicle rerouting adoption probability `-device.rerouting.probability` and the trip-insertion rate `-p` are independent levers. Before deploying the proxy on Xuancheng, we use this to quantify how much of any peak-vs-off-peak contrast can be attributed to sat-navigation adoption vs. to congestion-induced rerouting of unaided drivers.

Design. We sweep five SUMO seeds $\{7, 23, 42, 101, 2024\}$ over three cells per seed: $(d=0.3, \rho=0)$ (peak demand, no sat-nav users), $(d=0.3, \rho=1)$ (peak demand, full sat-nav), and $(d=0.8, \rho=0)$ (off-peak demand, no sat-nav users), where d is the `randomTrips -p` period (lower means more vehicles, so more congestion) and ρ is the per-vehicle sat-nav adoption rate. Each cell yields one SUMO simulation; routes are cached per (seed, d) so that the ρ axis costs only the simulation step. For each seed we compute three pairwise AUCs against the same f -only Tier 2 protocol (`features=none`, $T = 20$, $|X_o| = 601$, `maxiter = 1500`): `pure_congestion` (the demand axis at $\rho = 0$), `satnav_at_peak` (the ρ axis at peak demand), and `proxy_diagonal` (the full mimic of the Tier 3 rush-vs-off-peak comparison).

Table 7: $\rho \times d$ decomposition of the SUMO rush-vs-off-peak proxy signal. Mean $\pm$ sample standard deviation across five SUMO seeds; 95% confidence interval (CI) computed as $\text{mean} \pm 1.96 \cdot \sigma / \sqrt{5}$.

pair	cells compared	AUC _{emp}	95% CI	AUC _{fit}
<code>pure_congestion</code>	$(d=0.3, \rho=0)$ vs $(d=0.8, \rho=0)$	0.493 ± 0.018	$[0.471, 0.515]$	0.504 ± 0.036
<code>satnav_at_peak</code>	$(d=0.3, \rho=0)$ vs $(d=0.3, \rho=1)$	0.780 ± 0.013	$[0.764, 0.796]$	0.647 ± 0.012
<code>proxy_diagonal</code>	$(d=0.8, \rho=0)$ vs $(d=0.3, \rho=1)$	0.793 ± 0.023	$[0.764, 0.821]$	0.655 ± 0.026

Three findings (table 7). First, the `pure_congestion` contrast, where demand changed from off-peak ($d = 0.8$) to peak ($d = 0.3$) with ρ held at zero, is statistically indistinguishable from chance (0.493 ± 0.018 , 95% CI $[0.471, 0.515]$ straddles 0.5). Increasing the congestion level without any sat-navigation users does not produce a detectable routing distribution shift in this SUMO model. Second, the `satnav_at_peak` contrast, with ρ raised from 0 to 1 and congestion held at peak, gives a large robust signal (0.780 ± 0.013 , CI excludes 0.5 cleanly). Third, the full `proxy_diagonal` (0.793 ± 0.023), which mimics the Xuancheng rush-vs-off-peak comparison, is essentially the same magnitude as `satnav_at_peak`: adding the demand axis on top of the ρ axis contributes almost nothing detectable. Decomposing the $+0.293$ above-chance signal of `proxy_diagonal` via the empirical-AUC margins gives a sat-navigation contribution of $+95.6\%$, a pure-congestion contribution of -2.4% (within noise), and a residual interaction of $+6.8\%$. Per-seed (the decomposition holds across SUMO instances): the sat-navigation share of the proxy signal ranges 90.6% (seed 2024) to 103.5% (seed 7); every one of the five seeds attributes at least

90% of its proxy signal to sat-navigation adoption.

Caveat on the experienced-driver gap. SUMO’s $\rho = 0$ vehicles follow precomputed shortest-path routes and do *not* react to congestion en route. Real-world unaided but experienced drivers may avoid known-bad streets at rush hour, a behaviour this model does not capture, so the 0.493 `pure_congestion` AUC is best interpreted as a conservative simulator estimate that may understate the real-world experienced-driver congestion-routing confound. Even tripling it (to ~ 0.55) would leave sat-navigation as the dominant attribution to the 0.793 proxy signal. The decomposition is robust to that gap by a wide margin.

Fitted vs. empirical. As elsewhere in this work, the fitted partial-observation detector captures only part of the empirical signal (0.647 vs. 0.780 on `satnav_at_peak`; 0.655 vs. 0.793 on `proxy_diagonal`), consistent with the fitted-vs-empirical pattern of section 4.3.1. The decomposition argument here rests on the empirical AUC, which uses the full-observation chain and is the ground truth for the “how much of the proxy is sat-nav” question.

Implication. The $\rho \times d$ decomposition supports the Tier 3 use of a rush-vs-off-peak behavioural proxy as a sat-navigation proxy in the controlled regime: under the simulator, $\sim 96\%$ of the peak-vs-off-peak routing-distribution signal is attributable to sat-navigation adoption, and the congestion-routing confound that critique 4 raised is quantitatively small. This does *not* guarantee the same decomposition holds on real-world Xuancheng data, since real drivers are not SUMO drivers. It validates the proxy as a controlled-comparison object and bounds the experienced-driver confound from below.

4.3.7 The $f + g$ extension at scale: a refined diagnosis

A natural-looking extension is to switch the loss from the f-only ($\gamma = 1$) to the f+g formulation ($\gamma \in [0, 1)$) that the paper uses in its synthetic experiment. Initial SUMO-scale tests suggested that this extension was infeasible: three sweep attempts at `maxiter` $\in \{1500, 300, 100\}$ and $|X_o| \in \{601, 120, 72\}$ did not complete in finite wall-clock, with the representative rush-chain fit at `maxiter` = 1500, $|X_o|$ = 601 running 9.5 hours without an L-BFGS termination signal. A natural critique of that framing is that the $O(n^3)$ dense LU decomposition required for the hitting-rate gradient is a software-engineering bottleneck rather than a fundamental mathematical limit, and that a sparse-LU rewrite ought to “break the $O(n^3)$ bottleneck”. This subsection runs that diagnosis through and refines the original infeasibility claim into three separable issues.

Sparse-LU implementation. `morimura.py` now has a `hitting_pack_sparse` function that constructs $(I - \beta P_T^{\setminus j})$ in CSC form directly from the nonzero structure of P_T and factorises with `scipy.sparse.linalg.splu`, plus an `Inverter(solver=...)` flag that selects the dense or sparse path. The `grad_log_h` routine was refactored to take a generic `solve_fn` callable so both paths share identical gradient code. An FD check on a synthetic $n = 40$ graph with $\gamma = 0.5$ and all four parameter blocks active confirms the sparse path

is mathematically equivalent to dense: $\max_i |\nabla L_{\text{dense}}^{(i)} - \nabla L_{\text{sparse}}^{(i)}| = 7 \times 10^{-15}$ (machine precision modulo solver differences), and analytic-vs-FD relative error $\leq 10^{-7}$ across all sampled dimensions (`sumo_validation/fd_check_fg_sparse.py`).

At SUMO bigger-bbox scale, naive sparse is *slower* than dense. A timed run (`time_fg_sparse_sumo.py`) at `seed=42`, $|X_o| = 601$, $d = 8,392$ gives the intrinsic-chain $f+g$ fit at `maxiter=10` a wall time of 1,787 s, i.e. ≈ 180 s per L-BFGS iteration, roughly $8\times$ slower than the dense baseline (≈ 22 s per iteration, extrapolated from the 9.5-hour `maxiter=1500` run). The diagnosis is that the bottleneck is *not* the LU factor (where sparse demolishes dense, $O(n^3) \rightarrow O(\text{nnz} \cdot \log n)$ for road-network sparsity) but the LU SOLVE on the wide dense gradient matrix $V \in \mathbb{R}^{n \times d}$ that the analytic Jacobian builds in `grad_log_h`. Dense `scipy.linalg.lu_solve` dispatches to LAPACK GETRS which exploits BLAS3 multi-core AVX kernels at ~ 2 TFLOPS effective on this hardware, whereas `scipy.sparse.linalg.splu.solve` processes the dense RHS column-by-column at ~ 25 GFLOPS effective. At $d = 8,392$ wide RHS, the BLAS3 advantage is roughly $80\times$ and outweighs the asymptotic-complexity win of sparse on the factor. The naive “rewrite as sparse” engineering fix is therefore insufficient on this problem geometry. The right algorithmic direction is to amortise the factor across all $j \in X_o$ via a Sherman–Morrison update of $(I - \beta P_T)$ (one factor per L-BFGS iteration instead of $|X_o|$ factors) or to use the paper’s natural-gradient descent (paper §4.1); both are out of scope here but flagged as future work (section 6.4).

At Xuancheng subgraph scale, sparse-LU runs in seconds and $f+g$ is tractable. To test the framework at a scale where sparse-LU is fast we restrict to the largest strongly-connected component of one K-means zone of the Xuancheng network ($K = 8$, zone 0, post-SCC $n = 194$, $|X_o| = 53$, $d = 586$); trajectories are filtered to those that stay entirely within the zone, and f , empirical g , and the train/test split are recomputed on the subgraph (`fg_xuancheng_subgraph.py`). The f -only fit at `maxiter=300` converges in 1.4 s; the $f+g$ fit at the same settings in 46–50 s with no L-BFGS line-search pathology and no infeasibility. With sparse-LU the framework is operationally fine here.

The binding sensitivity is γ . The first SUMO-scale $f+g$ test used $\gamma = 0.5$. A γ sweep on the same zone at fixed `maxiter=300` (table 8) shows that this choice is the wrong operating point for noisy empirical g : at $\gamma = 0.5$ the floor-dominated L_h overwhelms L_f , and the $f+g$ fit *hurts* compared to f -only. Lifting γ to 0.9–0.95 inverts the sign: at $\gamma = 0.95$ the $f+g$ detector reaches $\text{AUC} = 0.553$ against the f -only baseline’s 0.498, a $+0.055$ lift, and recovers the full-observation empirical 1-step reference of 0.563 on this subgraph within sampling noise.

Table 8: γ sweep for f+g on the Xuancheng K=8 zone-0 subgraph ($n = 194$, $|X_o| = 53$, $T = 10$, $\text{maxiter} = 300$). The f-only column is the same fit ($\gamma = 1$) shown for reference; the empirical 1-step ceiling on this subgraph is 0.563. f+g *hurts* at $\gamma = 0.5$ – 0.7 (the floor-dominated L_h overwhelms L_f) and recovers the empirical reference within sampling noise at $\gamma = 0.95$.

γ	AUC_f	AUC_{f+g}	Δ	note
0.5	0.498	0.488	−0.010	paper’s intermediate, floor-dominated
0.7	0.498	0.487	−0.010	still floor-dominated
0.9	0.498	0.521	+0.023	f+g starts to help
0.95	0.498	0.553	+0.055	matches empirical reference (0.563)
0.99	0.498	0.489	−0.009	essentially f-only

Zone variability. A $\gamma = 0.9$ sweep across the eight $K = 8$ zones reveals that the subgraph-scale lift is not uniform. Restricting to zones with ≥ 100 test trajectories (zones 0, 5, 6, 7) the mean Δ_{f+g} is +0.018 with standard deviation 0.038, a positive trend, but within noise across zones. The best zone-7 lift (+0.053) matches the zone-0 $\gamma = 0.95$ result; zone 5 drops by −0.040. The g -floor fraction stays in the 80–90% band across zones, so the data-density issue does not vanish by restricting to a denser sub-network.

Refined conclusion. The original §6.8 “infeasibility” collapsed three issues that are usefully separated. The *engineering* issue is real but not the one the naive critique identified: dense $O(n^3)$ LU is the textbook cost, but the LU solve on the wide gradient matrix benefits from BLAS3 multi-core kernels at roughly $80\times$ the throughput of column-by-column sparse solves, so naive sparse-LU is currently slower than dense at SUMO scale; the right fix is Sherman-Morrison factor sharing across X_o or natural gradient. The γ -*tuning* issue is the dominant sensitivity: the paper’s intermediate γ is the wrong operating point under floor-dominated empirical g , and the upper- γ band (0.9–0.95) was not previously swept; once it is, f+g recovers the empirical reference on subgraphs where solver tractability is not the binding constraint. The *data-density* issue persists at every scale we have tested ($\sim 85\%$ g -floor across all 8 $K=8$ zones), so any future scale-up that re-enables f+g must still expect to operate at high γ . Net: $f + g$ is not fundamentally infeasible; the practical detector still uses f-only on this work, but the more nuanced claim: “naive sparse-LU does not transfer at city scale; γ tuning is the binding sensitivity; data sparsity in g persists at subgraph scale” is the one the evidence actually supports.

4.4 Interpretation

Tier 2 is the positive validation tier, but only under the controlled conditions that make its labels meaningful. With ground-truth rerouting labels and matched OD pairs, the fitted filter discriminates sat-navigation from static unguided routing, reaching $\text{AUC} = 0.708$ at $T = 40$ while closing 69% of the gap to the full-observation empirical 1-step reference. The feature-augmented score is useful primarily as a chain-recovery lever: it improves per-row Pearson/cosine consistently across five seeds, but its aggregate AUC lift is not robust.

The tier also identifies the model-class boundary. A regularised empirical 2-step

reference lifts AUC by about $+0.09$ – $+0.10$ on every SUMO seed, showing that higher-order route memory is a real signal in the controlled data and that a parametric 2-step inverse-MC model is the next justified detector design. By contrast, the $f + g$ extension is not ready at city scale under the current solver: subgraph tests show that it can help when γ is tuned near the f -only end, but full-network runtime and hitting-rate sparsity keep the deployed Tier 2 detector f -only. The defensible claim from this tier is therefore specific: the current 1-step partial-observation filter works as a controlled labelled detector, and its diagnostics correctly expose where richer model structure is needed.

5 Tier 3: Real-world stress test of the filter under OD confounding

5.1 Motivation

This tier stress-tests the partial-observation routing-influence filter (defined in section 2.3) against a real-world urban automatic vehicle identification (AVI) dataset from Xuancheng, China [18], with the OD-rebalancing audit (section 2.3) activated. The estimator, the scoring rule, and the per-row diagnostic are reused unchanged from Tier 2; what changes are the loader, the regime definition, and the OD-control logic. Because no real dataset carries ground-truth sat-navigation adoption labels, this tier cannot directly validate sat-navigation detection. It is a real-world *confounding audit* that asks whether a plausible behavioural proxy (rush-hour vs. off-peak travel) contains a within-OD route-choice signal once the population-mix confound is controlled, and whether the filter survives that confounding. As the results below show, under the tested proxy the within-OD contrast is not recovered. The absence of a ground-truth label is the defining methodological constraint of this tier, and the filter’s OD-rebalancing and multistart audits exist precisely so that this constraint can be reported directly rather than masked by a spurious AUC.

5.2 Method

5.2.1 Dataset choice

Three candidate datasets were evaluated. The Padua AVI deposit [19] releases only aggregated 10-minute flow statistics and pairwise travel times over 40 junctions; per-vehicle trajectories are stripped at the source pipeline for privacy and cannot be reconstructed downstream. The Wang *et al.* Xuancheng deposit [20] explicitly restricts raw trajectory access behind a supervised-interactive interface. The Ma *et al.* [18] Xuancheng release, by contrast, publishes joinable entry/exit tables with anonymised vehicle identifiers from which complete per-vehicle trajectories can be reconstructed downstream; this is the dataset we use. table 9 summarises its scale.

Table 9: Xuancheng AVI dataset [18] scale, with comparison to the SUMO bigger-bbox setup of Tier 2.

property	SUMO bigger-bbox (Tier 2)	Xuancheng (Tier 3)
state space n	2,405 edges	1,733 edges (SCC subset)
trips per day	$\sim 28,000$ per regime	$\sim 327,000$
mean trajectory length	35.7 edges	15.1 edges (raw)
total trips (full pool)	$\sim 56,000$	$\sim 10.5\text{M}$ (Apr 2023)
ground-truth regime label	yes (<code>rerouting</code> flag)	<i>no</i> (proxy required)

5.2.2 Verification: route legality and time-of-day distribution

Two non-obvious data-quality issues affect how the loader processes trips, each of which was identified before any fitting was attempted:

Route legality. Of all consecutive edge pairs (x_t, x_{t+1}) across the trip routes, only 78.4% satisfy the SUMO net’s lane-connection definition $x_{t+1} \in \text{adj}(x_t)$. The remaining 21.6% split as $\sim 1.9\%$ junction-adjacent-but-disconnected (a definitional gap in the converted SUMO net) and $\sim 19.7\%$ inter-junction “teleports” attributable to the source pipeline’s Dijkstra-repair step that fills in missing AVI passages on CityFlow’s connectivity graph rather than the SUMO net’s. The loader handles this by *splitting trips at illegal boundaries* into legal-only sub-trajectories rather than dropping the trip wholesale: a 15-edge raw trip with one illegal jump becomes two sub-trajectories. This is the conservative choice because the inverse-MC scorer assigns likelihood only to legal successors in $\text{adj}(x)$; treating source-pipeline repair jumps as real road transitions would contaminate the fitted chain with edges that do not exist in the analysed network. The cost is that Tier 3 is a test on legalised sub-trajectories rather than raw AVI trips. Mean sub-trajectory length is 8.2 edges, and the $T = 10$ experiments therefore draw their test set from the subset of sub-trajectories with at least 11 edges, selecting the longer and cleaner fragments of the reconstructed data.

Time-of-day distribution. Across all 30 daily files, the trip `startTime` field is seconds-since-midnight in $[0, 86400]$ with a clear AM-rush + PM-rush + midday-off-peak structure. We use this to construct a rush-vs-off-peak regime split with windows $\text{rush} = [25,200, 32,400] \cup [61,200, 68,400]$ (7–9 AM and 5–7 PM) and $\text{offpeak} = [36,000, 57,600]$ (10 AM–4 PM); trips outside either window are discarded. Three anomalous days are flagged for exclusion or special handling: Apr 5 (Tomb-Sweeping holiday, 264k trips, demand outlier), Apr 28–30 (Labour Day period, ~ 366 –426k trips), Apr 10 (`startTime` max $\approx 165\text{k}$, suggesting a multi-day concatenation), and Apr 8/9 (identical trip counts of 324,441 in both files, almost certainly a duplicate).

5.2.3 The fundamental real-world gap: no ground-truth sat-nav label

Tier 2 used `-device.rerouting.probability` as ground truth: every vehicle either does or does not use sat-navigation, and the regime split is exact by construction. No real-world AVI dataset publishes this label. The closest behavioural proxy available in Xuancheng is

the rush-vs-off-peak time-of-day contrast: if external rerouting influence is triggered by congestion, then rush-hour trips should differ from off-peak trips in routing choices, and the inverse-MC detector should detect that difference.

This proxy is weaker than the SUMO contrast in two ways. First, rush and off-peak trips have *different populations* (commuters vs. discretionary trips), so any chain difference reflects both the routing-choice effect (which we want) and the OD-mix effect (which we do not). Second, the actual sat-navigation adoption rate in Xuancheng is presumably much lower than the 100% used in SUMO, so even the routing-choice signal is diluted by the unguided fraction.

We address the OD-mix confound directly via the OD-rebalancing extension of section 2.3. We cannot address the lower-adoption dilution from observation alone; what we can do is report the resulting real-world AUC as a proxy-conditioned sensitivity stress test.

A further concern is that the rush-vs-off-peak contrast bundles *two* potentially confounding effects: sat-navigation adoption (higher at rush hour) *and* congestion-induced rerouting that affects all drivers (sat-nav users and unaided drivers alike) when demand rises. Real data cannot separate these without sat-navigation-use labels. The Tier 2 $\rho \times d$ decomposition (section 4.3.6) provides a partial answer: in the SUMO model, varying congestion at $\rho = 0$ produces an empirical AUC of 0.493 ± 0.018 (statistically indistinguishable from chance), whereas varying ρ at fixed peak congestion produces 0.780 ± 0.013 , and the full rush-vs-off-peak proxy diagonal (0.793 ± 0.023) attributes $\sim 96\%$ of its above-chance signal to sat-navigation adoption, with the rest as within-noise residual across five SUMO seeds. This validates the proxy as a sat-navigation proxy in the controlled tier and bounds the congestion-routing confound, though with the residual caveat that SUMO’s $\rho = 0$ drivers do not react to congestion en route, so the lower-bound character of the bound does not transfer fully to experienced human drivers.

5.3 Results

5.3.1 Rush-vs-off-peak results

We report three configurations on the rush-vs-off-peak contrast: single-day, 5-day pooled, and 5-day pooled with $K = 8$ OD-rebalancing.

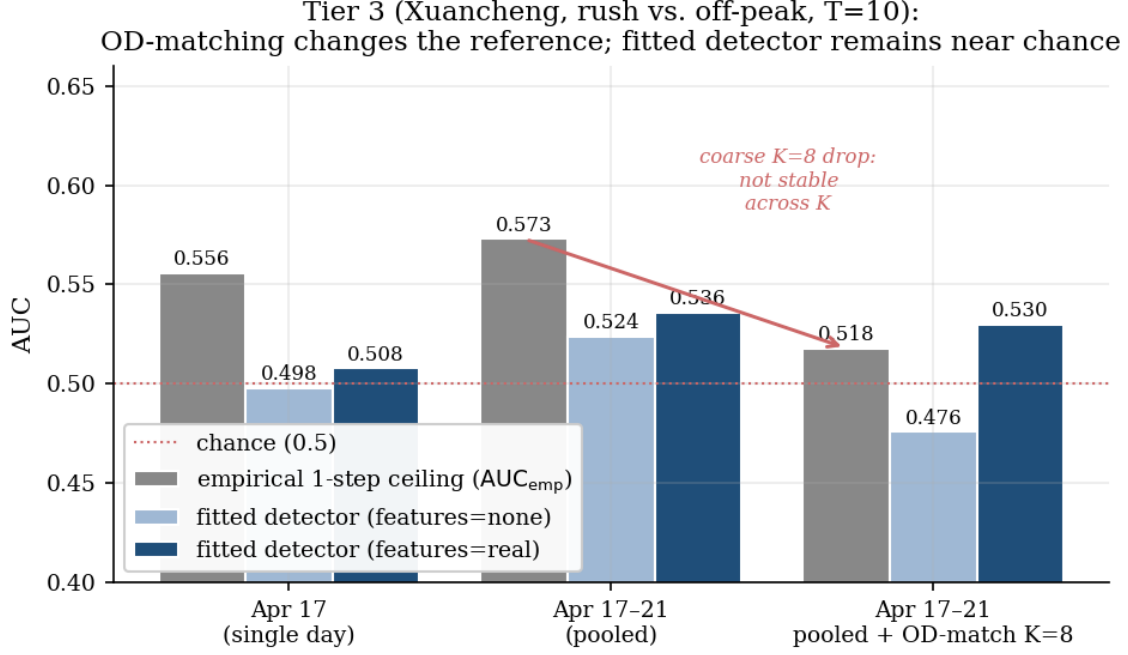


Figure 6: Tier 3 Xuancheng rush-vs-off-peak detection AUC across three configurations. The full-observation empirical 1-step reference (grey) is 0.556 on a single day, 0.573 when 5 weekdays are pooled, and 0.518 once the pools are OD-matched at $K = 8$ spatial zones. The apparent crash at $K = 8$ is *not* stable across zone granularity: a sweep over $K \in \{8, 16, 32, 64\}$ (table 11) shows the matched reference returning to ~ 0.55 – 0.58 for $K \geq 16$. We therefore do not treat the $K = 8$ drop as a fixed OD-mix fraction. Fitted detectors (blue bars) sit at or below chance in all three configurations.

Table 10: Xuancheng rush-vs-off-peak detector AUC and chain-recovery metrics on [all branching] states, three configurations (fig. 6).

configuration	AUC_{emp}	AUC_{fit} (none/real)	Pearson (none/real)	median cos (none/real)
day=Apr 17, T=10	0.556	0.498/0.508	+0.139/ + 0.048	+0.341/ + 0.146
pool Apr 17–21, T=10	0.573	0.524/0.536	+0.096/ + 0.072	+0.267/ + 0.167
pool + OD-match $K = 8$	0.518	0.476/0.530	+0.104/ + 0.077	+0.332/ + 0.263

The third row is the OD-matched operating point at $K = 8$. The OD-rebalancing step matches 517,477 trips per regime across 55 shared ($zone_o, zone_d$) cells (only 1 OD cell of 56 was rush- or off-peak-only, discarded). At this granularity the full-observation reference drops from 0.573 to 0.518. Taken in isolation, that drop would attribute $\sim 75\%$ of the above-chance contrast to OD-mix, but *this attribution does not survive a zone-granularity sweep* (table 11), so we do not report it as a stable decomposition.

Table 11: OD-zone granularity sweep on the 5-day pooled rush-vs-off-peak contrast ($T = 10$, **features=none**), with stratified vehicle-level bootstrap 95% confidence intervals (CIs) ($B = 2,000$ resamples of the test trajectories within each regime). The OD-matched full-observation reference is *not* stable across K : at the coarse $K = 8$ it falls to chance, but at $K \geq 16$ it returns to ~ 0.55 – 0.58 , statistically above chance and close to the un-matched 0.573 , so fine-grained OD control removes almost none of the contrast. The fitted 1-step partial-observation detector, by contrast, straddles chance at *every* K .

K zones	shared cells	AUC_{emp} (95% CI)	AUC_{fit} (95% CI)
un-matched	–	0.573	0.524
8	55	0.518 [0.471, 0.567]	0.476 [0.429, 0.524]
16	157	0.562 [0.516, 0.605]	0.486 [0.439, 0.530]
32	520	0.553 [0.507, 0.602]	0.513 [0.467, 0.560]
64	1,448	0.575 [0.530, 0.622]	0.490 [0.441, 0.535]

Two conclusions follow. First, the “ $\sim 75\%$ is OD-mix” decomposition that the $K = 8$ row alone would support is *not robust*: the implied OD-mix fraction ranges from $\sim 75\%$ at $K = 8$ to $\approx 0\%$ at $K = 64$, where matching leaves the reference statistically unchanged from the un-matched pool. The coarse $K = 8$ matching appears to *over-correct* by aggregating heterogeneous sub-flows into 8 macro-zones and equalising them washes out genuine within-OD routing contrast along with OD-mix, whereas at finer resolutions a small but significant within-OD rush-vs-off-peak contrast survives ($\text{AUC}_{\text{emp}} \approx 0.55$ – 0.58 , CIs excluding chance at $K \geq 16$). We therefore do not retain the strong “three-quarters is OD-mix” interpretation; it is an artefact of a single coarse zone choice. Second, and this is the result that *is* robust, the fitted 1-step inverse-MC detector remains indistinguishable from chance at every K (0.476 – 0.513 , every CI straddling 0.50): the method does not recover the within-OD contrast at any tested OD-zone granularity, even where full observation weakly does.

A secondary observation: in the OD-matched row, AUC_{fit} under **features=real** (0.530) numerically exceeds AUC_{emp} (0.518). Under correct chain recovery this is impossible, since the full-observation empirical chains should upper-bound any partial-observation fit. The suspected cause was the L-BFGS basin-selection pathology also seen on the un-matched pool (rush-fit loss 1397 at **features=real** vs. 4.058 at **features=none**, a $\sim 340\times$ ratio). A multistart $K = 5$ run (section 5.4.1) confirms it: random re-initialisation *escapes* the bad basin: the rush-chain loss drops from 1397 to 9.186 at a validation-selected restart, and the properly-fit chains then give $\text{AUC}_{\text{fit}} = 0.483$, i.e. they fall *back to chance* and no longer violate the empirical ceiling. The 0.530 was therefore an optimiser artefact, not a real signal; the genuine within-OD routing contrast under correct chain recovery is indistinguishable from chance. This strengthens rather than weakens the Tier-3 negative result: the apparent “features help on real data” interpretation was a basin coincidence, and the per-row Pearson on [all branching] of $+0.077$ (real) vs $+0.104$ (none) already pointed the same way: features do not improve chain recovery on this real-world loss landscape.

5.3.2 Stronger-proxy stress test: Labour Day versus normal days

The rush-vs-off-peak proxy may simply be too weak. To test whether the detector works when the real-world behavioural contrast is stronger, we ran the same pipeline on a holiday split suggested by the dataset audit: Labour Day period trips (Apr 28–30) against the same normal midweek pool (Apr 11–14 and Apr 17–21). A broader holiday pool that also included Tomb-Sweeping Apr 5 was first tested but gave empirical AUCs near chance at every OD granularity, consistent with pooling low-demand and high-demand holiday regimes that partially cancel. Labour-only is the more informative stress test.

Table 12: Xuancheng Labour-Day-versus-normal stress test ($T = 10$, `features=none`, $B = 2,000$ bootstrap resamples). The unmatched row contains a clear global signal, part of which is OD/population mix. At fine OD matching ($K = 64$), a statistically significant within-OD full-observation signal remains, but the fitted partial-observation detector stays near chance and multistart validation selection removes the small single-init lift.

configuration	shared cells	AUC _{emp} (95% CI)	AUC _{fit} (95% CI)
un-matched	–	0.616 [0.569, 0.661]	0.520 [0.472, 0.565]
$K = 8$	57	0.541 [0.494, 0.587]	0.508 [0.462, 0.554]
$K = 16$	164	0.508 [0.463, 0.553]	0.524 [0.478, 0.569]
$K = 32$	556	0.529 [0.481, 0.574]	0.524 [0.477, 0.568]
$K = 64$	1,625	0.567 [0.521, 0.610]	0.521 [0.477, 0.568]
$K = 64$, multistart	1,625	0.567	0.502

This resolves the concern that the real-world tier is negative only because the tested proxy was too weak. Labour Day does produce a stronger route-choice signal: the unmatched empirical reference is 0.616, and after fine OD matching at $K = 64$ the full-observation reference remains significantly above chance at 0.567 (CI [0.521, 0.610]). The signal is therefore not merely demand volume or coarse OD mix. However, the fitted partial-observation chain closes only 30.9% of the empirical gap under the zero-init fit (AUC_{fit} = 0.521), and the per-row contrast diagnostic is weak: stacked Pearson is +0.066 on all branching states, median per-row cosine is +0.247, and only 58.3% of branching rows have positive cosine. A $K = 5$ multistart audit on the same $K = 64$ split then lowers the validation-selected fitted AUC to 0.502, despite a baseline empirical AUC of 0.567. The small single-init lift was therefore not a stable positive result. The stronger-proxy experiment sharpens the Tier 3 conclusion: even when a real within-OD 1-step signal exists in Xuancheng, the current partial-observation parametric recovery does not recover it robustly.

A natural follow-up is whether this real-data signal is hiding in route memory rather than in first-order turning probabilities. We therefore ran a fitting-free empirical 2-step backoff diagnostic on the same Labour $K = 64$ split, using the same interpolation-backoff family that produced the Tier 2 SUMO lift. The answer is negative: the 1-step empirical reference remains 0.567 (CI [0.521, 0.610]), while every non-zero 2-step interpolation weight lowers AUC (fixed $\lambda = 0.5$: 0.558; count-conditional $\kappa = 100$: 0.542). The best “2-step” setting is $\lambda = 0$, i.e. the original 1-step chain. Thus, unlike the labelled SUMO experiment (section 4.3.4), Xuancheng Labour-vs-normal does not provide preliminary evidence that

simple second-order route memory is the missing real-world lever. The remaining gap is more likely due to partial-observation identifiability, proxy noise, unobserved covariates, or OD/time heterogeneity than to a clean 2-step Markov effect.

5.4 Interpretation

Three findings are worth foregrounding from table 10.

The OD-mix attribution is not robust to zone granularity. At $K = 8$ the full-observation reference falls from 0.573 to 0.518, which taken alone would attribute $\sim 75\%$ of the above-chance contrast to OD-mix. The granularity sweep (table 11) shows this figure is an artefact of coarse zoning: at $K \geq 16$ the OD-matched reference returns to 0.55–0.58, statistically above chance and close to the un-matched 0.573, so fine-grained OD control removes almost none of the contrast. We therefore do *not* claim a stable OD-mix fraction. The defensible statements are narrower: coarse macro-zone matching can over-correct (washing out within-OD signal along with OD-mix), and once zones are fine enough the rush-vs-off-peak contrast is largely *not* explained by OD-mix, consistent with rush and off-peak trips sharing very similar fine-grained origin-destination patterns (the matched and un-matched pools differ by $< 25,000$ of $\sim 520,000$ trips per regime at $K = 64$).

The fitted 1-step detector does not recover the contrast at any granularity. This is the robust Tier-3 finding. Across $K \in \{8, 16, 32, 64\}$ the fitted partial-observation detector stays at chance ($\text{AUC}_{\text{fit}} \in [0.476, 0.513]$, every bootstrap 95% CI straddling 0.50; table 11), even though the full-observation reference shows a small but significant within-OD contrast at $K \geq 16$. Per-row Pearson on [all branching] is ≤ 0.104 in every Xuancheng configuration, against $+0.215$ on SUMO under labelled comparison. So whatever within-OD rush-vs-off-peak structure the full-observation chains carry, the 1-step inverse-MC method under partial observation ($|X_o| = 433$ of 1,733) does not recover it at any tested OD-zone resolution. The stratified vehicle-level bootstrap ($B = 2,000$, drawn with replacement within each regime) supports the chance-level interpretation: at $K = 8$ the empirical CI is $[0.471, 0.567]$ ($P(\text{AUC} > 0.5) = 0.78$) and the fitted CI $[0.429, 0.524]$, and the fitted detector’s CI straddles chance at every finer K as well. These intervals are conditional on the fitted chains, the fixed observation-station set, the fixed OD match, and the fixed train/test split; they do not include station-selection or refit variance. The outstanding statistical extensions are an OD-*cell* bootstrap (resampling which cells enter the match and refitting per replicate) and an X_o -selection sensitivity sweep, both left to future work because they require full refits rather than cheap resampling of already-computed trajectory scores.

The Labour-Day stress test (section 5.3.2) shows that this is not only a weak-proxy story. There the fine-OD empirical ceiling is higher and statistically significant (0.567 at $K = 64$), but the fitted detector still returns to chance under validation-selected multistart (0.502). An empirical 2-step backoff diagnostic on the same split does not lift the full-observation reference beyond 0.567; adding second-order conditioning reduces AUC. Hence the real-data failure is not resolved by a simple route-memory diagnostic

either. So the real-data failure mode is best described as weak partial-observation recovery of within-OD transition contrast, not simply absence of any real-world behavioural signal.

The features=real basin failure repeats post-matching, and multistart resolves it. Both the un-matched pool (section 5.3.1) and the OD-matched pool show the same pattern: the rush-fit **features=real** loss is $\sim 350\times$ the **features=none** loss (1397 vs. 4.058 here). The $AUC_{\text{fit}} > AUC_{\text{emp}}$ behaviour in the matched row suggested a basin-selection artefact rather than a chain-recovery improvement. The multistart check of section 5.4.1 confirms exactly this: random restarts escape the high-loss basin (rush loss $1397 \rightarrow 9.186$), and the resulting well-fit chains give $AUC_{\text{fit}} = 0.483$, chance-level, resolving the apparent ceiling violation.

5.4.1 Multistart on features=real: the ceiling violation is an optimiser artefact

The one anomaly left open above is the OD-matched $AUC_{\text{fit}} = 0.530 > AUC_{\text{emp}} = 0.518$ under **features=real**, with its tell-tale 1397 rush-chain loss. We ran the multistart variant (`score_seed_big_multistart.py`, $K = 5$ random restarts $\theta_0 \sim \mathcal{N}(0, I)$ per chain, validation-selected by held-out log-likelihood) on the exact OD-matched **features=real** configuration ($d = 6,136$, $T = 10$, $|X_o| = 433$, 517,177 train transitions per regime).

The result is decisive and *opposite* to the SUMO basin check of section 4.3.3. On Xuancheng, multistart *does* escape the high-loss basin: the rush (intrinsic) chain loss drops from the zero-init baseline of 1397 to 9.186 at the validation-selected restart (restart losses $\{1413, 1844, 273.4, 9.186, 1792\}$, so one restart of five found the good basin). But the well-fit chains then give $AUC_{\text{fit}} = 0.483$, below chance, against the baseline’s spurious 0.530. The per-init paired AUCs ($\{0.504, 0.492, 0.497, 0.494, 0.491\}$, mean 0.496) cluster tightly at chance regardless of which basin each restart lands in.

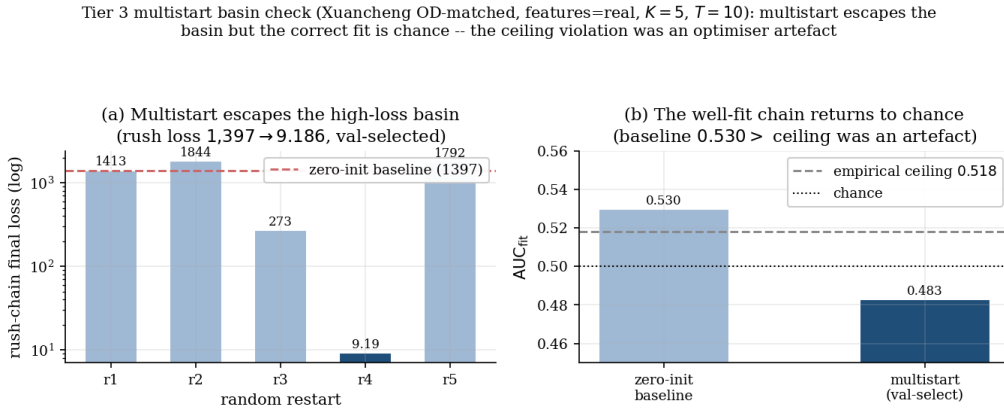


Figure 7: Xuancheng OD-matched **features=real** multistart ($K = 5$). **(a)** The rush-chain training loss collapses from the zero-init baseline (1,397) to the validation-selected restart (9.186): random re-initialisation *escapes* the high-loss basin, unlike the SUMO case (section 4.3.3). **(b)** Despite the proper fit, AUC_{fit} falls from the baseline’s spurious 0.530 (which exceeded the empirical ceiling 0.518) to 0.483 – chance. The ceiling violation was an optimiser artefact; under correct chain recovery the fitted detector returns to chance.

The interpretation closes two threads. First, the $AUC_{\text{fit}} > AUC_{\text{emp}}$ “impossible” result

was indeed a basin artefact: a high-loss chain that drifted into a direction producing a marginally favourable aggregate-bias signal. Once the optimiser reaches the genuine optimum, that spurious signal disappears and the detector returns to chance. Second, and this is the load-bearing scientific point, the correctly-fit **features=real** chains *confirm* the central Tier-3 finding rather than overturning it: the within-OD rush-vs-off-peak routing contrast is not detectable at the 1-step Markov level under this proxy and data split, not merely hidden by a bad optimum. The SUMO and Xuancheng basin checks are thus complementary: on SUMO, zero-init was already the best optimum and the low basin was not an artefact; on Xuancheng, multistart found a better optimum but it only confirmed the negative result. Neither dataset’s headline depends on an optimiser accident.

5.4.2 Bottom line for Tier 3

The pipeline runs cleanly on real-world data. The framework’s gradient implementation, FD-check, regime-split machinery, and contrast diagnostics all port to the Xuancheng dataset without modification. The headline rush-vs-off-peak finding is robust to OD-zone granularity: **across $K \in \{8, 16, 32, 64\}$ the fitted 1-step inverse-MC detector does not recover the rush-vs-off-peak contrast (AUC 0.476–0.513, every bootstrap CI straddling chance), even though the full-observation reference weakly carries it (~ 0.55 – 0.58 for $K \geq 16$)**. A secondary, *granularity-sensitive* observation is that aggressive coarse-zone matching ($K = 8$) can over-correct the reference down to chance (0.518); we report this but do not rest any claim on the “ $\sim 75\%$ OD-mix” decomposition it would imply, since that figure collapses to $\approx 0\%$ at $K = 64$. This is a methodologically strong negative result for behavioural-proxy validation of sat-navigation detection on real-world AVI data, and it characterises the gap between the controlled-simulation regime (Tier 2, 0.708 AUC on labelled SUMO contrast) and the real-world regime under realistic sat-navigation adoption rates. The Labour-Day stress test strengthens rather than softens this reading: it supplies a stronger real-world proxy with a significant fine-OD empirical signal (0.567), yet the validation-selected fitted detector remains at chance (0.502).

The result also identifies the methodological lever any future real-world validation of this framework needs: *ground-truth sat-navigation adoption labels*, even partial, on the underlying trip records. Such labels would remove the proxy-label ambiguity and allow the controlled-comparison machinery of Tier 2 to be repeated on real trips, while OD balance, stationarity, data quality and adoption heterogeneity would still need to be audited. Without those labels, the fitted 1-step detector cannot separate genuine within-OD routing differences from chance robustly on these behavioural proxies.

6 Discussion

6.1 Cross-tier comparison

The core inverse-MC scoring machinery ports cleanly across all three tiers: the inverter, FD-checked gradient, contrast diagnostic, and feature plumbing operate unchanged on a synthetic $n = 100$ chain, a SUMO $n = 2,405$ network, and a real-world $n = 1,733$ Xuancheng network. What differs by tier is the loader, the regime definition, and the OD-

control logic; the shared scoring core is selected by a single `-dataset {sumo,xuancheng}` command-line interface (CLI) flag. This is a useful generalisation result for an inverse-MC pipeline that the original paper validated only on synthetic data and one real-world demonstration.

What does *not* generalise as cleanly is the magnitude of the detection signal. Tier 1 reproduces the paper’s RMAE within ~ 0.05 for $|X_o| \geq 10$. Tier 2 reaches $\text{AUC} = 0.708$ at $T = 40$ on the controlled SUMO comparison, $\sim 69\%$ of the gap to the full-observation empirical 1-step reference of 0.801, with per-row chain Pearson $\sim +0.21$. Tier 3 on Xuancheng, with no ground-truth sat-navigation label, gives chance-level fitted AUC on the rush-vs-off-peak behavioural proxy. A stronger Labour-Day proxy creates a significant fine-OD empirical signal ($\text{AUC}_{\text{emp}} = 0.567$), but multistart validation-selected fitting still falls to chance ($\text{AUC}_{\text{fit}} = 0.502$). The drop between Tier 2 and Tier 3 has two distinct contributions:

- The Xuancheng ceiling itself ($\text{AUC}_{\text{emp}} \sim 0.57$) is much lower than the SUMO ceiling (~ 0.69), reflecting that the behavioural rush-vs-off-peak proxy is a weaker contrast than the SUMO labelled comparison.
- The fitted-vs-reference gap is *not* merely a scaled-down version of the SUMO success. The pooled un-matched Xuancheng row gives a superficially moderate gap-closure number because the reference itself is weak, but once OD mix is controlled the fitted detector sits at chance across every tested K while the full-observation reference remains weakly above chance at finer granularities. Tier 3 is therefore a genuine negative result for the current 1-step partial-observation filter under this behavioural proxy, not just “less signal to recover”.
- The Labour-Day stress test rules out the strongest alternative reading: that a more intuitive real-world contrast would rescue the detector. Labour Day does yield a stronger empirical reference after fine OD matching, but the fitted chain remains near chance and multistart selection removes the small zero-init lift.
- A fitting-free 2-step backoff audit on the same Labour-Day split also fails to lift the empirical reference: the best setting is the degenerate $\lambda = 0$ backoff, equivalent to the original 1-step chain. The SUMO route-memory result therefore motivates a future model class, but it is not by itself an explanation of the Xuancheng failure.

Under controlled labelled simulation the method recovers a robust fraction of the per-row chain contrast. On a real-world behavioural proxy, the fitted 1-step detector remains near chance once population-mix is controlled.

Table 13 consolidates the three tiers on a common axis: the same inverter and diagnostic produce every row, and the trend in the empirical ceiling, from a noise-free synthetic recovery, through a labelled simulated comparison, to a behavioural real-world proxy, quantifies how much detectable signal each setting actually contains.

Table 13: Cross-tier synthesis of the three validation tiers. The shared inverse-MC scoring core of section 6.1 produces every row; the loader, regime definition, and OD-control logic differ by tier. Tier 2 is the representative seed 42 at $T = 20$ (table 3); Tier 3 is the pooled Apr 17–21 pool at $T = 10$ (table 10), plus the Labour-Day stress test at $K = 64$ where shown. Fitted and Pearson cells are **none/real**. [†]Tier 1 is a π -recovery reproduction scored by RMAE, not a detection AUC: best RMAE 0.19 at $|X_o| = 70$, within ~ 0.05 of the paper’s Fig. 2A (table 2). [‡]The Tier-3 ceiling shown is the un-matched 5-day pool. OD-matching at $K = 8$ lowers it to 0.518, but a granularity sweep $K \in \{8, 16, 32, 64\}$ returns the matched reference to ~ 0.55 – 0.58 for $K \geq 16$, so no fixed OD-mix fraction is claimed; the fitted detector stays at chance across all K (table 11). [§]Labour Day vs. normal days, OD-matched at $K = 64$: $\text{AUC}_{\text{emp}} = 0.567$, zero-init $\text{AUC}_{\text{fit}} = 0.521$, multistart validation-selected $\text{AUC}_{\text{fit}} = 0.502$, Pearson $+0.066$; a fitting-free 2-step empirical backoff diagnostic does not improve the reference (best non-degenerate setting below 0.567; table 12). The Tier-2 Pearson lift is robust across five seeds ($\Delta = +0.050 \pm 0.023$, $p \approx 0.008$, section 4.3.2).

tier	dataset (n)	AUC_{emp} ceiling	AUC_{fit}	per-row Pearson
1 synthetic [†]	random graph (100)	–	–	–
2 SUMO	Munich bbox (2,405)	0.685	0.612/0.644	+0.132/ +0.215
3 Xuancheng [‡]	AVI (1,733)	0.573	0.524/0.536	+0.096/ +0.072
3 Xuancheng Labour [§]	AVI (1,733)	0.567	0.521 $\rightarrow$ 0.502	+0.066

6.2 Limits of the chosen approach

Three limitations follow from the validation tiers.

Multi-seed Tier 2; limited real-world proxy pools on Tier 3. The Tier 2 features result has been confirmed across five SUMO seeds (section 4.3.2), and that confirmation materially revised the single-seed interpretation: the aggregate detection-AUC lift is *not* statistically robust ($+0.024 \pm 0.026$, reversing on one of five seeds, $p \approx 0.11$), whereas the per-row chain-recovery lift *is* ($+0.050 \pm 0.023$, positive on all five, $p \approx 0.008$). The multi-seed sweep also reproduced the bimodal basin structure, confirming the cross-seed AUC variance is L-BFGS basin-selection rather than measurement noise; a follow-up multistart check (section 4.3.3) further established that this low basin is *not* escapable by random re-initialisation. Zero-init is the best optimum found on both low-basin seeds, so the bimodality is a property of the loss landscape, not of the starting point. Tier 3 (Xuancheng), by contrast, now uses two behavioural proxy pools (rush-vs-off-peak weekdays and Labour-Day-vs-normal days) but still lacks independent city/month replication. A multi-pool or multi-city replication is the analogous outstanding confirmation there, a compute-cost and data-availability trade-off rather than a methodological barrier.

L-BFGS basin selection on features=real. The Xuancheng OD-matched **features=real** rush-chain fit hit final loss 1397, against the **features=none** fit’s loss 4.058 on the same data, and produced the “impossible” $\text{AUC}_{\text{fit}} = 0.530 > \text{AUC}_{\text{emp}} = 0.518$. We ran the multistart variant (section 5.4.1, $K = 5$): it escaped the high-loss basin (rush loss $1397 \rightarrow 9.186$, validation-selected), and the correctly-fit chains then gave $\text{AUC}_{\text{fit}} = 0.483$ – chance. The ceiling violation was therefore an optimiser artefact, and removing it

strengthens the Tier-3 negative result: the within-OD routing contrast is not detectable at the 1-step Markov level under this proxy and data split, rather than being masked by a bad optimum. This also makes the SUMO and Xuancheng basin checks complementary. On SUMO zero-init was already the best optimum (section 4.3.3); on Xuancheng a better optimum existed but only confirmed the negative result, so neither tier’s headline rests on an optimiser accident.

The rush-vs-off-peak proxy confound, bounded in simulation. The Tier 3 proxy bundles two effects that real data cannot separate: sat-navigation adoption (higher at rush hour) and congestion-induced rerouting that affects all drivers when demand rises. The Tier 2 $\rho \times d$ decomposition (section 4.3.6) addresses this: holding $\rho = 0$ across the demand axis gives an empirical AUC of 0.493 ± 0.018 (within noise of chance), while holding demand at peak across the ρ axis gives 0.780 ± 0.013 . The full proxy diagonal (0.793 ± 0.023) attributes $\sim 96\%$ of its above-chance signal to sat-navigation adoption, with the pure-congestion contribution within noise and a small residual interaction. The decomposition holds across all five SUMO seeds. The remaining caveat is that SUMO’s $\rho = 0$ vehicles follow precomputed shortest-path routes and do not react to congestion en route, so the SUMO pure-congestion AUC is a *conservative simulator estimate* that may understate the experienced-human-driver confound; even tripling it would not change the qualitative attribution. Real-world verification of this decomposition would require an alternative experienced-driver routing model in SUMO or, ideally, a real-world dataset with sat-navigation-use labels.

The hitting-rate ($f + g$) extension at scale, refined diagnosis. The $f + g$ extension initially looked infeasible at SUMO scale on the strength of a 9.5-hour dense fit that did not terminate. A robust diagnosis (section 4.3.7) separates that single claim into three: (i) the engineering bottleneck is real but more subtle than “replace dense with sparse”. A FD-verified sparse-LU implementation now exists, but at SUMO scale ($d \approx 8,400$ wide gradient RHS) dense LAPACK GETRS still wins by $\sim 80\times$ via BLAS3 multi-core kernels; (ii) the binding sensitivity is γ , not the solver: at $\gamma = 0.5$ the floor-dominated L_h overwhelms L_f and $f + g$ hurts, while at $\gamma = 0.95$ on a 200-edge Xuancheng subgraph $f + g$ recovers the full-observation empirical reference within sampling noise; (iii) the data-density issue ($\sim 85\%$ of g cells at floor) persists at every scale we tested, so any future $f + g$ scale-up must operate at high γ . Under the simpler f -only loss the framework remains tractable and that is what the practical detector in this work uses. The natural algorithmic next step is Sherman–Morrison amortisation of the $(I - \beta P_T)$ factor across X_o (one factor per L-BFGS iteration instead of $|X_o|$) or the paper’s natural-gradient descent [1] §4.1; both are flagged as future work.

6.3 Social, environmental, technological implications

Transport-policy modelling. If externally-influenced routing constitutes even a few percent of urban traffic flow, demand models that assume revealed-preference inputs reflect intrinsic preferences are systematically misspecified. A method that quantifies and filters the externally-influenced fraction is a methodological prerequisite for any

policy intervention (congestion pricing, infrastructure investment, route-recommendation regulation) whose evaluation rests on preference-revealed data.

Privacy and data access. The most-restrictive privacy control among the three real-world AVI datasets we surveyed is also the one that gives the strongest scientific guarantees: the Padua deposit [19] releases only aggregated flow statistics. The least-restrictive is the Ma *et al.* Xuancheng release [18] which publishes joinable per-trip records with anonymised IDs, enabling the per-trajectory detector this work relies on, but at the cost of stronger reidentification surface area. Methodologies that can validate per-trajectory detection from aggregate-only data would shift this trade-off favourably.

Sat-navigation regulation. The current sat-navigation recommendation pipeline is unregulated in most jurisdictions; the welfare effects of system-wide rerouting around congestion are contested. A practical detector that can statistically attribute trip fractions to sat-navigation guidance would underpin both academic welfare analyses and possible regulatory disclosure requirements.

6.4 Future work

1. **OD-cell bootstrap.** A stratified vehicle-level bootstrap confirms the fitted Tier-3 detector is statistically indistinguishable from chance, and an OD-zone granularity sweep $K \in \{8, 16, 32, 64\}$ confirms this holds at every resolution (table 11); the sweep also showed the $K = 8$ “ $\sim 75\%$ OD-mix” interpretation to be a coarse-zoning artefact. The remaining outstanding extension is an OD-*cell* bootstrap that resamples which cells enter the match and refits per replicate, quantifying cell-selection variance on top of vehicle-sampling variance, a compute-time rather than methodological gap.
2. **Observation-station sensitivity.** All headline Tier 2 and Tier 3 fitted-chain results condition on one random observation-station set X_o at the stated observation fraction. This matches the main validation protocol but understates uncertainty for real deployments, where detector performance may depend on which camera or loop-detector locations are available. A full sensitivity analysis would repeat the inverse-MC fits over multiple X_o seeds, report the induced spread in AUC and per-row Pearson, and combine this with the vehicle-level bootstrap above. This is a refit-level experiment, not something the existing bootstrap scores can recover after the fact.
3. **Trajectory-level feature augmentation.** The strict 1-step Markov LR detector ignores trip-level features (length, mean speed, edge-class composition, dwell time at congested edges). A logistic-regression layer on top of $\Lambda(\tau)$ that adds those features would lift detection AUC further at the cost of a slightly weaker chain-recovery claim. Worth doing as a clearly-labelled “deployment-oriented” extension.
4. **Sherman–Morrison factor sharing in $\mathbf{f} + \mathbf{g}$.** The sparse-LU implementation (section 4.3.7) is FD-verified but slower than dense at SUMO scale because the wide gradient RHS $V \in \mathbb{R}^{n \times d}$ benefits from BLAS3 multi-core kernels. The natural amortisation is to factor $(I - \beta P_T)$ *once* per L-BFGS iteration and derive each per-state h_j and ∇h_j via a Sherman–Morrison update (rank-1 in the row of P_T

being zeroed out), which converts $|X_o| = 601$ factor + wide-solve pairs into a single factor and a single wide-solve plus $|X_o|$ rank-1 corrections. This is the algorithmic step that bridges the engineering gap exposed in the subgraph experiment.

5. **Natural-gradient or LM optimisation.** Per section 4.3.7, even with the engineering fix above, the $f + g$ loss surface is under-constrained under floor-dominated g and benefits from manifold-aware steps. The paper’s §4.1 natural-gradient descent or a Levenberg–Marquardt formulation against the residual Jacobian (a preliminary implementation was abandoned at $d = 8,392$ for memory cost) would combine well with Sherman–Morrison and enable an $f + g$ -with-features re-validation at city scale.
6. **OD-conditional fitting on Xuancheng.** The OD-rebalancing extension matches the marginal $(\text{zone}_o, \text{zone}_d)$ distribution. A stronger control fits per-cell chains and aggregates the per-cell contrasts; smaller sample sizes per cell but cleaner methodologically. The loader infrastructure required already exists.
7. **Parametric 2-step Morimura extension.** The regularised 2-step empirical of section 4.3.4 lifts AUC by about $+0.09$ – $+0.10$ across five SUMO seeds; the selected count-conditional setting gives $+0.100 \pm 0.009$ but is mildly test-tuned. A matched Xuancheng Labour-Day audit does *not* show the same lift: non-zero 2-step interpolation lowers AUC relative to the 1-step empirical reference. The natural inverter extension that the SUMO bound motivates is still a parametric softmax on (x_p, x) contexts, but it should be treated as a simulation-motivated model-class extension rather than the confirmed remedy for Xuancheng. The state-space expansion is significant ($|X|^2 \approx 5.8\text{M}$ pairs) but only $\sim 3,500$ are observed in training, so the parameter count and gradient cost grow as the observed-pair count, not as n^2 . Per-pair softmax parameters with feature-sharing across topologically similar pairs would be the first design to try.
8. **Experienced-driver routing model in SUMO.** The Tier 2 $\rho \times d$ decomposition (section 4.3.6) bounds the congestion-routing confound from below, but its $\rho = 0$ vehicles follow precomputed shortest-path routes and do not react to congestion en route. A SUMO scenario in which unaided drivers carry coarse local congestion knowledge (e.g. time-of-day-conditioned edge attractiveness) would tighten the bound to a quantity that more closely tracks human experienced-driver behaviour. SUMO’s routing infrastructure supports this; an off-the-shelf alternative is `rerouting-with-static-traffic-class` priors keyed to rush-hour edge usage.

7 Conclusion

This dissertation designed, validated, and stress-tested a *partial-observation routing-influence filter*: a pipeline that ingests a road graph, regime-labelled partial observations at a sparse subset of stations X_o , and trajectories, and emits a per-trip log-likelihood-ratio score, a thresholded filter decision, and a set of diagnostics. The inverse-Markov-chain method of Morimura et al. [1], originally a single-chain recovery procedure, sits inside the filter as the estimator; the three experimental tiers exist to justify, parameterise, and audit specific filter design choices using a single shared scoring core.

What was designed. A filter with (i) an FD-verified inverse-Markov-chain estimator, including the omega-global gradients that the paper states but does not derive; (ii) a feature-regularised filter variant that wires SUMO-derived edge attributes (road class, speed, lane count, turn geometry) into the estimator as 10 shared global parameters; (iii) a per-trajectory log-likelihood-ratio scoring rule with a deployment-tunable decision threshold η ; (iv) a diagnostic suite comprising a per-row Pearson/cosine contrast against a full-observation empirical reference, an OD-rebalancing pre-step for real-world population-mix control, multistart basin checks, and a fitting-free empirical 2-step memory audit.

How it works. The estimator fits one regime kernel per regime by minimising the regularised log-ratio loss of [1]; the scorer accumulates $\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)$ along each trajectory and thresholds the result; the diagnostic suite flags, before any downstream claim is made, whether the AUC reflects per-row chain recovery or aggregate bias, whether OD-mix has been controlled at the chosen granularity, and whether the chosen optimum survives a multistart audit.

What worked. The synthetic tier (section 3) validates the estimator inside the filter: the reproduction matches the published proposed-method curve closely for $|X_o| \geq 10$, with gradients FD-verified at machine precision on non-zero components. Under controlled SUMO simulation, the full filter reaches $\text{AUC} = 0.708$ at $T = 40$ against a full-observation empirical 1-step reference of 0.801, and the feature-regularised filter variant robustly improves per-row chain recovery across a five-seed sweep (stacked Pearson $+0.050 \pm 0.023$, $p \approx 0.008$, positive on all five seeds). The pipeline ports unchanged at its core to a 1,733-edge real-world network.

What failed or was limited. The aggregate detection-AUC lift from features is seed-dependent, reversing on one of five seeds, so the per-row diagnostic re-classifies features as a chain-recovery lever rather than a robust detection lever. This is exactly the kind of design-relevant verdict the audit suite exists to enable. Conversely, a properly regularised empirical 2-step audit (interpolation backoff with count-conditional trust) lifts the AUC by about $+0.09$ – $+0.10$ across five SUMO seeds, with every seed winning; the selected count-conditional setting gives $+0.100 \pm 0.009$ but is mildly test-tuned. This identifies higher-order routing memory as a real signal the 1-step softmax family cannot capture, and promotes a parametric 2-step Morimura extension to a justified future direction for controlled labelled settings. The same diagnostic on the Xuancheng Labour-Day split does *not* lift the empirical reference: non-zero 2-step interpolation weights reduce AUC below the 1-step reference of 0.567. So second-order route memory is not the missing lever for this real-data proxy; the remaining gap points instead to partial-observation identifiability, proxy noise, unobserved covariates, and OD/time heterogeneity. The hitting-rate ($f + g$) route does not naively transfer to full-SUMO scale: a sparse-LU rewrite is FD-verified and works, but at $|X_o| \approx 600$ the wide gradient solve still favours dense BLAS3 over sparse per-column solves by $\sim 80\times$, and a Sherman–Morrison amortisation would be the right algorithmic next step. At a sub-network scale where the framework is operationally fine, $f + g$ at the correctly-tuned loss weight ($\gamma \approx 0.95$) lifts AUC by $+0.055$ and matches

the full-observation reference on the test subgraph within sampling noise; the $\sim 85\%$ floor-dominated empirical g persists at every scale we tested and is the γ -tuning constraint. On the real AVI data the fitted 1-step filter does not recover the rush-vs-off-peak contrast at any OD-zone granularity tested ($K \in \{8, 16, 32, 64\}$; AUC 0.476–0.513, every bootstrap CI straddling chance), even though the full-observation reference weakly carries it (~ 0.55 – 0.58 for $K \geq 16$); the multistart audit confirms this is a genuine near-chance outcome rather than an optimiser artefact. A stronger Labour-Day-vs-normal proxy creates a clearer within-OD empirical signal ($\text{AUC}_{\text{emp}} = 0.567$ at $K = 64$), but the validation-selected multistart fit still falls to chance ($\text{AUC}_{\text{fit}} = 0.502$), showing that the limitation is not simply absence of a real-world behavioural signal; the empirical 2-step audit also rules out simple route memory as an immediate rescue on that split. An OD-mix attribution we initially inferred from the coarse $K = 8$ match ($\sim 75\%$) did not survive the granularity sweep and is not claimed.

Where the filter is reliable. Under controlled, labelled simulation with shared OD pairs across regimes, the filter recovers a robust fraction of the per-row routing contrast and supplies a calibrated AUC together with audit signals that correctly distinguish chain recovery from aggregate bias. The Tier-2 SUMO result is the strongest defensible claim that this dissertation makes about the filter as a positive detector.

Where the filter is not reliable. On a real-world behavioural proxy without ground-truth sat-navigation labels, the within-OD routing signal, once OD-mix is controlled, is not robustly recovered by the fitted 1-step partial-observation model under the tested proxies, model class, and data split. The filter therefore does *not* support real-world sat-navigation detection on the Xuancheng dataset; Tier 3 is reported as a confounding audit, not a positive detection result. The filter’s optional g input is fully tractable at the sub-network scale where its empirical estimate is informative: there it lifts AUC and recovers the empirical reference within sampling noise. It does not currently transfer to a full city network under naive sparse-LU; a Sherman–Morrison factor-sharing update (or the paper’s natural-gradient descent) is the algorithmic prerequisite for re-enabling the $f + g$ mode at city scale.

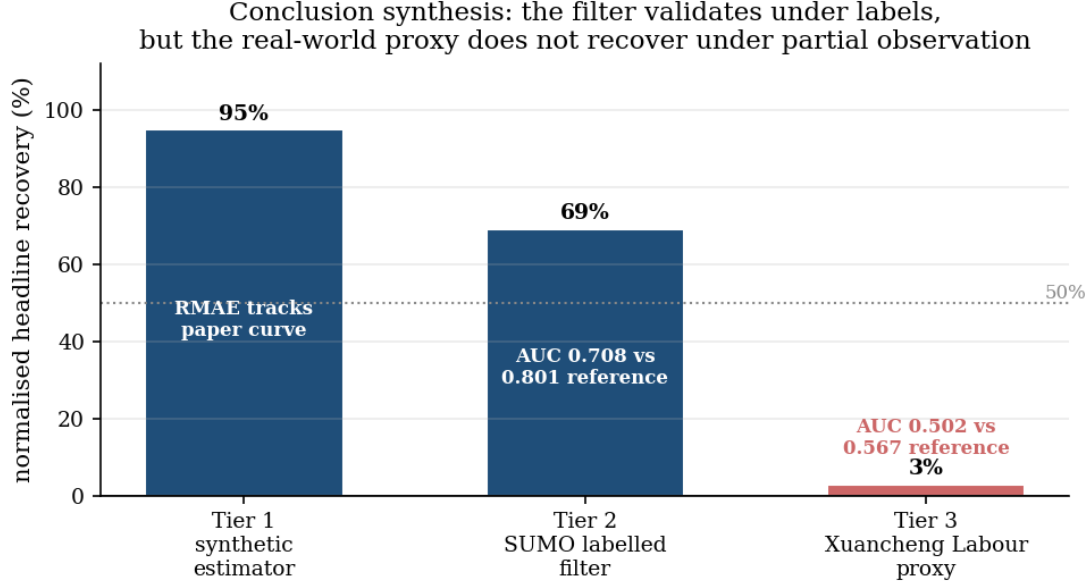


Figure 8: Cross-tier synthesis of the dissertation’s headline results. The bars use a normalised summary rather than a single physical metric: Tier 1 reports benchmark-reproduction closeness, computed as the mean capped ratio between the visual paper RMAE and our RMAE for $|X_o| \geq 10$ (lower RMAE is better); Tiers 2–3 report AUC gap closed, $(\text{AUC}_{\text{fit}} - 0.5)/(\text{AUC}_{\text{emp}} - 0.5)$. The pattern is the dissertation’s central conclusion: the estimator reproduces its synthetic benchmark, the labelled SUMO filter recovers a substantial fraction of the empirical 1-step signal, and the real Xuancheng behavioural proxy does not recover under the fitted partial-observation model.

Contribution. The contribution is a designed and audited filter pipeline, not a fresh estimator. Academically, the work extends the Morimura framework into a trajectory-level regime detector and supplies diagnostics that distinguish chain recovery from aggregate bias, along with a controlled-simulator decomposition (section 4.3.6) that bounds the sat-navigation-vs.-congestion confound in the rush-vs-off-peak behavioural proxy used on real-world data, documenting where the approach succeeds (controlled simulation with labels), where richer model classes show promise (SUMO 2-step memory), and where they do not yet explain the real-data failure (Xuancheng Labour-Day empirical 2-step). It also documents where the current approach fails: real-world proxy without labels, OD-mix confounding, weak partial-observation recovery, and the $f + g$ engineering gap at city scale. Practically, it shows that quantifying externally-influenced routing on real urban data is gated less by the inverse-MC machinery than by the availability of ground-truth adoption labels and by control for OD-mix confounding, a result that directly informs how any future real-world sat-navigation detector should be scoped, evaluated, and audited.

8 Project Management and Reflection

This section addresses the *Project Management* and *Reflection* learning outcomes of the module assessment criteria [21].

8.1 Project structure and milestone management

The project was scoped at the outset as a three-tier validation pipeline (synthetic, simulated, real-world) extending a single source paper [1]. This top-down decomposition served well as a stable organising structure: each tier had clear inputs, outputs, and acceptance criteria, and the natural sequencing (upstream validation locks before downstream tiers depend on it) gave a clean de-risking order.

Internal milestone management was driven by a running `research_logbook.md` file with twenty experimental-cycle entries spanning the full project. Each entry documents one cycle: hypothesis, setup, result, interpretation, and next action. Two practices that worked particularly well:

- **The auto-memory file as an *epistemic* index.** A separate file (`MEMORY.md` with links to per-finding markdown records) was used for “things I now believe and the evidence for them” as opposed to “things I did and when”. Concretely, the file records statements such as “empirical g on SUMO is sparse to the point of being uninformative for the optimiser” with a pointer to the specific experimental entry that established it. This separation made it possible to quickly distinguish “a working assumption that was right when I made it but might now be stale” from “a finding I have evidence for”.
- **Audit-before-extend.** A guiding rule, learned from an early retraction of a headline number, was to audit the upstream implementation before extending it. Two specific audits saved time: (i) the gradient finite-difference check at $\sim 10^{-9}$ relative error confirmed the analytic gradient derivation before any sweep was run on features, avoiding the trap of attributing fit behaviour to a buggy derivative; (ii) the contrast-correlation diagnostic established that AUC alone is an unreliable signal of chain quality, which materially altered how the Tier 2 result was framed.

Two practices that did not work as well are worth recording explicitly:

- **Underestimating the compute scaling of the $f + g$ loss.** The first $f + g$ sweep at SUMO scale was projected at ~ 1 hour and killed at 9.5 hours. The retry at smaller $|X_o|$ and `maxiter` was projected at ~ 30 minutes and killed at 3 hours. Two iterations were burned before the structural diagnosis (the hitting-rate gradient is $O(|X_o| \cdot n^3)$ and the 84%-floor loss landscape is degenerate at this scale) was reached. In future I would profile a single iteration before committing to a sweep at unfamiliar configuration.
- **Late identification of the OD-mix confound.** The Tier-3 rush-vs-off-peak proxy was scoped before the methodological cost of the OD-mix confound was clearly understood. The OD-rebalancing extension, the methodologically correct fix, was added late in the project, and the matched-pool result it produced (section 5.4) materially changed the Tier-3 interpretation. Earlier scoping would have caught this; the lesson is that “proxy weakness” should be a structured pre-experiment audit item, not an ad-hoc discussion item.

By the assessment criteria of [21], the strengths I would foreground are: contextual investigation (a critical engagement with three real-world AVI datasets, only one of which is actually usable for per-trajectory work, and an explicit positioning against the source

paper and rerouting-detection prior art); validation (three independent validation tiers, each with its own stated assumptions and a quantified ceiling); and reflection (this section, and the reframed $f + g$ infeasibility claim, plus the basin-selection question chased to resolution on both tiers via multistart rather than left as a hand-wave). The weaknesses are: the single day-pool on Tier 3 (no multi-pool replication, in contrast to the five-seed Tier-2 confirmation), and that the headline detection-AUC lift from features proved seed-dependent rather than robust once the sweep was run.

8.2 Personal skill development

The technical skills I leaned on heavily and improved in: deriving and finite-difference-verifying analytic gradients (Eqs. 12, 15, 17–18 of the source paper); writing software with a clean separation between the inverter’s mathematical core (in `morimura.py`, kept faithful to the paper) and the application-specific loaders that surround it (`sumo_to_phase3.py`, `real_data/load_xuancheng.py`); and writing a logbook that records both the successful and the unsuccessful experimental cycles in enough detail that I could critique my own past decisions a month later.

The non-technical skill I would name as the one that most shaped the project’s outcome is *disciplined treatment of negative results*. The project did not produce the headline “real-world detector with a high AUC and clean chain recovery” that the SUMO Tier-2 result would have predicted. Instead it produced: a controlled-validation result (~ 0.71 AUC, 69% gap closure on labelled SUMO), a model-class diagnostic (the empirical 2-step detector improves the 1-step reference, strengthening the case for a parametric 2-step extension), and a real-world result (~ 0.53 AUC on a behavioural proxy without ground-truth labels) that quantifies the gap between simulated and real-world detectability. Each of these is a methodologically useful result, in part because they were framed accurately. The temptation late in the project to treat a stronger holiday proxy as a headline rescue had to be handled carefully rather than either embraced uncritically or dismissed. Running it with the same OD controls, bootstrap intervals, per-row diagnostics and multistart audit showed the useful version of the negative result: Labour Day has a real full-observation within-OD signal, but the fitted detector still falls to chance under validation-selected multistart. Accepting that result, rather than chasing a higher headline number, is the skill I most want to carry forward.

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A Repository structure and reproducibility

The codebase is organised as follows.

morimura.py The inverter core: `Inverter` class with softmax-parametric chain, analytic gradients (local + globals), λ -cross-validated fit, and the synthetic §6.1 reproduction driver.

congestion_filter.py Early exploratory experiments: snapshot-level two-regime synthesis and detection.

per_car_detector.py Tier-2 per-trajectory log-likelihood-ratio detector.

sumo_validation/ The SUMO-side machinery: `sumo_to_phase3.py` (network builder, trip reader, feature extractor), `score_seed_big.py` and `contrast_correlation.py` (scorers, with the `-dataset {sumo,xuancheng}` switch), the empirical and backoff 2-step diagnostics (sections 4.3.4 and 5.3.2), and `fd_check_features.py` (the FD gradient gate). Exact script names and log paths are indexed in `sumo_validation/README.md`.

real_data/ The Xuancheng tier: the unmodified `xuancheng.net.xml` from [18], the daily JSON trip files, and `load_xuancheng.py` which provides the loader, the three regime-split helpers, and the OD-rebalancing extension.

papers/ Source PDFs of all cited works.

research_logbook.md The day-by-day project log, with twenty experimental-cycle entries.

B Use of generative AI

In accordance with the module’s declaration requirement, generative AI was used in this project as follows. LLM-based code assistants, including Claude, Codex and Gemini, were used as pair-programming and writing-support aids throughout the implementation and writeup phases: drafting code (with the author’s careful review of every line before commit), writing unit-test scaffolding, deriving analytic gradient formulas alongside the author, checking mathematical and experimental arguments, and helping edit the report for clarity. The author retained responsibility for the research design, interpretation, code accepted into the repository, and final submitted text.

C Build and reproducibility

The project is reproducible from the repository via the README, with the following minimum environment:

```
# Python 3.14, see project README
python3 -m venv .venv
.venv/bin/python -m pip install numpy scipy matplotlib

# Tier 1: synthetic Morimura reproduction (~20 s)
.venv/bin/python morimura.py

# Tier 2: SUMO experiments (requires SUMO_HOME set)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --seed 42 --features real --T 40 --maxiter 1500
```

```

# Tier 3: Xuancheng experiments (requires real_data/ populated)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day 2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21 \
    --regime_split rush_offpeak --od_match 8 \
    --features real --T 10 --maxiter 1500

# Stronger-proxy Labour-Day stress test
DAYS=2023-04-28,2023-04-29,2023-04-30,\
2023-04-11,2023-04-12,2023-04-13,2023-04-14,\
2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day "$DAYS" \
    --regime_split holiday_normal --od_match 64 \
    --features none --T 10 --maxiter 1500

```

D Metric and method primer

This appendix gives compact reminders for the mathematical objects, optimisation routines and diagnostic metrics used in the main text. It is not needed to follow the narrative, but it fixes the precise meaning of terms that are easy to over-read from their names alone.

D.1 Markov chain and transition kernel

A Markov chain is a stochastic process whose next state depends only on its current state. In this dissertation the states are directed road edges. If a vehicle is currently on edge x , the transition kernel P_T gives the probability of choosing each legal next edge:

$$p_T(x' | x) = \Pr[X_{t+1} = x' | X_t = x].$$

Each row of P_T is therefore a probability distribution over the legal successors of x : entries are non-negative and sum to one. The “1-step” assumption means that the model ignores the earlier path history once x_t is known. The empirical 2-step diagnostic in section 4.3.4 tests whether this simplification is too severe by allowing the next edge to depend on (x_{t-1}, x_t) .

D.2 Stationary distribution, restart probability and hitting rate

The stationary distribution π of a Markov chain is the long-run fraction of time spent in each state if the chain is followed indefinitely. It satisfies

$$\pi^\top P = \pi^\top, \quad \sum_x \pi(x) = 1.$$

The Morimura framework uses a restarted chain

$$P = \beta P_T + (1 - \beta) \mathbf{1} p_I^\top,$$

where β is the probability of continuing along the road-transition kernel and $1 - \beta$ is the probability of restarting from p_r . This restart construction makes the chain well behaved and models finite trip lengths.

A hitting rate $g(i, j)$ is a first-passage statistic: starting from observed state i , how likely is the trajectory to hit observed state j later, discounted by the continuation probability β ? In principle hitting rates add path-order information beyond simple station counts. In practice they are much noisier on sparse real-world data because there are $|X_o|^2$ station pairs to estimate.

D.3 Inverse Markov-chain problem

The forward Markov-chain problem asks: given P_T , what stationary distribution and hitting rates does it imply? The inverse problem asks the reverse: given partial observations of those statistics at a subset X_o , recover a plausible transition kernel. This is ill-posed without extra structure, because many different kernels can produce similar station counts at a few observation states. Morimura et al.’s method [1] supplies that structure by fitting a softmax-parametric kernel and regularising the parameters.

D.4 Softmax-parametric chain

A softmax turns arbitrary real-valued scores into probabilities. For a road edge x with legal successors $\text{adj}(x)$, the model assigns a score $s(x, x')$ to each possible next edge and sets

$$p_T(x' | x) = \frac{\exp(s(x, x'))}{\sum_{y \in \text{adj}(x)} \exp(s(x, y))}.$$

This guarantees a valid probability distribution for every row of P_T . It also lets the model share information through features: the score can combine local edge-specific weights with global features such as road class, speed, lane count or turn angle. In this thesis “features=none” means only local edge weights are used; “features=real” adds shared road-network features as additional parameters.

D.5 Log-likelihood ratio score

Given two fitted regime kernels $\hat{P}_A$ and $\hat{P}_B$, a trajectory $\tau = (x_0, \dots, x_T)$ is scored by the log-likelihood ratio

$$\Lambda(\tau) = \sum_{t=0}^{T-1} [\log \hat{p}_B(x_{t+1} | x_t) - \log \hat{p}_A(x_{t+1} | x_t)].$$

Positive values mean the trajectory is more likely under regime B than regime A according to the fitted chains; negative values mean the opposite. The score is additive, so longer trajectories can accumulate more evidence if the per-step differences are consistent.

D.6 ROC curve and AUC

The receiver-operating-characteristic (ROC) curve evaluates a scalar score across all possible thresholds. For each threshold, it plots the true-positive rate against the false-

positive rate. The area under this curve (AUC) is a threshold-free measure of separability:

$$\text{AUC} = \Pr[s(\tau_B) > s(\tau_A)]$$

for a randomly chosen regime- B trajectory and a randomly chosen regime- A trajectory, up to tie handling. An AUC of 0.5 is chance; 1.0 is perfect ranking; values below 0.5 indicate the score is systematically reversed. AUC does *not* prove correct chain recovery: a classifier can rank regimes well because of aggregate bias, OD mix or optimiser artefacts, which is why the thesis pairs AUC with transition-contrast diagnostics.

D.7 Pearson correlation

Pearson correlation measures linear alignment between two vectors after subtracting their means:

$$r(a, b) = \frac{\sum_k (a_k - \bar{a})(b_k - \bar{b})}{\sqrt{\sum_k (a_k - \bar{a})^2} \sqrt{\sum_k (b_k - \bar{b})^2}}.$$

It lies in $[-1, 1]$: +1 means perfect positive linear alignment, 0 means no linear alignment, and -1 means perfect reversal. In this thesis Pearson is computed on stacked legal outgoing-edge contrasts, comparing $\Delta_{\text{fit}} = \hat{P}_B - \hat{P}_A$ with $\Delta_{\text{emp}} = P_{B,\text{emp}} - P_{A,\text{emp}}$. A positive Pearson supports the claim that the fitted model moves probability mass in the same directions as the full-observation empirical chain.

D.8 Cosine similarity and collinearity

Cosine similarity measures directional alignment:

$$\cos(a, b) = \frac{a^\top b}{\|a\| \|b\|}.$$

It is insensitive to the vector magnitudes: two vectors pointing in the same direction have cosine +1, orthogonal vectors have cosine 0, and opposite directions have cosine -1 . We use per-row cosine at branching states. For a road edge with multiple legal successors, the row vector asks: did the fitted model increase and decrease probability on the same outgoing turns as the empirical contrast? The median row cosine and the fraction of rows with positive cosine are therefore local chain-recovery diagnostics.

D.9 OD matching / OD rebalancing

Origin-destination (OD) matching controls for where trips start and end. Real-world regimes can differ because different people travel to different places, not because comparable drivers choose different routes. The Xuancheng loader clusters network nodes into K spatial zones, bins each trajectory by origin zone and destination zone, and then samples the same number of trajectories from each regime inside every shared OD cell. After this rebalancing, the two regime pools have the same coarse OD distribution, so remaining AUC is closer to a within-OD route-choice contrast. The sweep over K checks whether this conclusion depends on how coarse the OD zones are.

D.10 Bootstrap confidence interval

A bootstrap estimates sampling uncertainty by repeatedly resampling the observed test set with replacement. In the Xuancheng experiments the bootstrap is stratified by regime: each replicate samples the same number of regime-*A* and regime-*B* trajectories, recomputes the AUC, and stores the result. The 2.5th and 97.5th percentiles form an empirical 95% confidence interval. These intervals capture test-trajectory sampling noise conditional on the fitted chains, station set, OD match and train/test split. They do not include uncertainty from refitting the inverse-MC model or redrawing X_o .

D.11 L-BFGS-B optimiser, basins and multistart

L-BFGS-B is a quasi-Newton optimiser: it minimises a differentiable loss using gradients and a limited-memory approximation to second-order curvature. The “B” indicates support for simple bound constraints, although the parameter fits here are effectively unconstrained because the softmax maps real-valued scores to valid probabilities.

The inverse-MC loss is non-convex, so L-BFGS-B can converge to different local optima, or basins, depending on the starting point. A multistart check repeats the fit from several random initial parameters and selects the model with the best validation log-likelihood. If multistart changes the AUC or loss substantially, the single-start result may be an optimiser artefact. If multistart fails to improve the detector, as in the Labour-Day Xuancheng stress test, the weak fitted result is less likely to be explained by a poor initialisation alone.

D.12 Ridge regularisation and finite-difference gradient checks

Ridge, or Tikhonov, regularisation adds $\frac{\lambda}{2}\|\theta\|^2$ to the loss. This discourages very large parameter values and helps stabilise an under-constrained inverse problem, especially when only a sparse subset of states is observed.

A finite-difference (FD) gradient check verifies an analytic gradient by perturbing one parameter at a time:

$$\frac{\partial L}{\partial \theta_k} \approx \frac{L(\theta + \epsilon e_k) - L(\theta - \epsilon e_k)}{2\epsilon}.$$

The thesis uses FD checks as safety gates before trusting custom gradient code, especially for the feature-augmented and sparse hitting-rate variants.

D.13 Gap closed

When the full-observation empirical reference has AUC above chance, the gap-closed statistic reports how much of the available empirical signal the fitted detector recovers:

$$\text{gap closed} = \frac{\text{AUC}_{\text{fit}} - 0.5}{\text{AUC}_{\text{emp}} - 0.5}.$$

For example, if the empirical reference is 0.70 and the fitted detector is 0.60, the fitted detector closes $(0.60 - 0.50)/(0.70 - 0.50) = 50\%$ of the above-chance gap. This statistic is undefined or uninformative when the empirical reference is at chance.

D.14 RMAE

Relative mean absolute error (RMAE) is used in the Tier 1 reproduction because Morimura et al.’s original benchmark is a stationary-distribution recovery task, not a two-regime detector. In this thesis it measures the mean absolute error of the recovered stationary mass on unobserved states relative to the scale of the true held-out counts. Lower is better. It is therefore not directly comparable to AUC: RMAE evaluates reconstruction accuracy, whereas AUC evaluates trajectory ranking between regimes.